# Higher Education Expansion and the Rural–Urban Gap in Secondary Schooling and Labor Outcomes: Evidence from China's 1999 Reform

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**Abstract**

This study examines how China's 1999 expansion of university admissions reshaped rural and urban students' academic high-school choices and, in turn, rural–urban gaps in schooling and employment. Treating the nationwide expansion of tertiary capacity as an exogenous shock, we use microdata from the 2015 1% Population Sample Survey of China to estimate difference-in-differences models with county and cohort fixed effects. Relative to comparable urban cohorts, rural students exposed to the reform were 2.9 percentage points more likely to complete the academic high school track and accumulated an additional 0.26 years of schooling. The expansion also reallocated rural workers away from low-skill employment by 5.1 percentage points. These findings indicate a strong indirect channel operating through upper-secondary choices, while underscoring the need for complementary investments in rural upper-secondary schools and local non-farm labor markets. The evidence is consistent with changes in expectations and application behavior at the upper-secondary transition: national tertiary expansion increased the perceived attainability of the academic track for rural cohorts, inducing adjustments in continuation decisions.

**Keywords:** Higher Education Expansion; Rural–Urban Disparities; Secondary Schooling; Human Capital; Labor Market Outcomes
**JEL Classification:** I21; I24; J24; O15; R23

# 1 Introduction

In 1999, China launched a decade-long expansion of higher education, with annual university enrollment rising from roughly one million to over 6.3 million within ten years.[1] Policy documents and contemporaneous commentary emphasized two broad motivations: alleviating short-run employment pressures and, more importantly, accelerating human-capital accumulation to support structural transformation and productivity growth (Che and Zhang 2018; Feng and Xia 2022). The reform directly expanded tertiary places through rapid seat creation and admission-quota adjustments, while also changing the informational environment and perceived returns associated with pursuing college. In a setting where basic education was nearly universal but university slots remained scarce, and where the economy was increasingly skill-intensive, the expansion made the college pathway more attainable and more legible to families and students along the schooling sequence.

In our setting, upper-secondary education in China is explicitly tracked into academic high schools and vocational high schools.[2] Academic places are allocated competitively via exam-based admissions, and scarcity has historically been acute, especially outside large cities. Prior to the reform, rural students were under-represented in tertiary channels and faced higher direct and indirect costs of continuation, which translated into lower entry into the academic track and lower progression to college for those who did enter (Li et al. 2015). Related work shows that rural students generally lag behind urban students in academic performance and face substantial dropout risks, with schooling costs especially binding among rural junior- and upper-secondary students, consistent with binding financial and information constraints (Yi et al. 2012). In practice, even academically qualified rural students often opted for the vocational track because of liquidity constraints, information gaps, and the premium placed on near-term employment stability.

Secondary schooling is the key stage linking primary schooling to the pathways into university and skilled employment. Empirical work shows that expanding and improving this stage raises cohorts' human capital, boosts productivity, and can improve intergenerational mobility (Hanushek and Wößmann 2006; Psacharopoulos and Patrinos 2004). Classic work in human-capital economics emphasizes why investments in the middle grades matter for earnings and growth, through skill accumulation, selection, and spillovers (Becker 1964; Mincer 1974; Lucas 1988; Krueger and Lindahl 2001). Consistent with this argument, evidence shows that when reforms lower costs or open up more places at key

[1] Enrollment counts are from the *Educational Statistics Yearbook of China*. See Che and Zhang (2018) and Feng and Xia (2022) for syntheses of the reform's timing, scale, and contemporaneous debates.

[2] We use "academic high school" to refer to the general academic track that feeds into the national college entrance examination, and "vocational high school" to denote upper-secondary vocational programs.

transition points, students respond: exam participation rises, academic-track entry increases, and persistence improves (Duflo 2001; Lucas and Mbiti 2012; Brudevold-Newman 2021). In China specifically, primary and lower-secondary enrollment and completion rose rapidly—especially after the 1986 nine-year Compulsory Education Law—but these gains did not translate into comparably smooth progression into the academic high school track, making upper-secondary tracking a key bottleneck.

Motivated by this institutional setting, we study whether the 1999 university expansion generated upstream responses at the upper-secondary track-choice stage and, in turn, narrowed rural–urban gaps in schooling and employment. The key question is not only how many college seats the reform added, but whether the tertiary-tier shock changed behavior at the entry margin to academic high school—where information frictions, risk, and local capacity constraints are first-order—and whether these responses differed for rural and urban students. This perspective aligns with recent work emphasizing that education systems should be analyzed as sequences, tracing how shocks at one tier propagate through the rest of the system (Jensen 2010; Iranzo and Peri 2009; Lucas 1988).

A key mechanism is belief and incentive updating. The 1999 expansion made the college pathway not only effectively more attainable, but also more legible: as admissions rose quickly and persistently, families could more clearly infer higher returns to the academic route and a higher likelihood of eventually reaching tertiary education. This mechanism parallels evidence from Texas's "Top 10%" rule, where a simple and transparent automatic-admission cutoff induced strategic sorting around the eligibility threshold (Cullen, Long, and Reback 2013) and substantially changed flagship enrollment choices for students near the margin (Daugherty, Martorell, and McFarlin 2014). In China's pre-reform environment, rural students' perceived disadvantage reflected a broader bundle of constraints than exam performance alone: limited access to information and guidance, weaker school resources and preparation, tighter household liquidity and higher perceived downside risk of "betting" on the academic track, and local capacity and competition at the upper-secondary transition. With historically very low rural admission rates, many rural households rationally treated tertiary education as outside their realistic opportunity set and developed downward-biased beliefs about the feasibility and payoff of the college path. The visible, sustained rise in university admissions therefore provided a credible signal to information-constrained families, facilitating belief updating. Together, these forces raised perceived net benefits of the academic track and altered upper-secondary track choices, inducing academic-track take-up among rural students with latent ability.

Empirically, we employ a quasi-experimental difference-in-differences design that compares cohorts whose upper-secondary track choices were plausibly affected by the 1999 reform to slightly older cohorts whose trajectories were largely predetermined by the time

of the expansion (Roth et al. 2023; Miller 2023; Sun and Shapiro 2022). Although the policy was announced and implemented nationally, its upstream effects on secondary education should operate primarily through cohorts still in junior secondary school at the time, before the academic-versus-vocational high school decision is made. We use this cohort-based exposure variation to estimate how the expansion shifted upper-secondary track choice and subsequent schooling and employment outcomes, with particular attention to within-county rural–urban gaps.

The distributional consequences of the reform depend crucially on how different groups were positioned before 1999. In particular, rural students faced structural disadvantages in school quality, exam-preparation resources, and family support (Bian and Li 2012; Downey, von Hippel, and Broh 2004), which raised the marginal costs of continued schooling and lowered baseline progression into and beyond upper secondary (Downey 2008; Erten and Keskin 2018; Fish 2010). Precisely because starting points differed, the same nationwide expansion could generate systematically different behavioral responses and gains for rural versus urban students. Guided by this framework, we use rural–urban contrasts as our primary approach to identifying distributional effects: after controlling for common county- and cohort-level shocks, changes in within-county rural–urban gaps across cohorts more directly reflect the reform's relative impact across groups.

A key assumption of our design is that rural and urban individuals born in the same county were, on average, exposed to the same local shocks and policies while growing up. This allows us to absorb county and cohort fixed effects that flexibly control for county- and cohort-specific shocks that are common to rural and urban residents, such as county-level policy shifts, macroeconomic fluctuations, and changes in educational resources (Wooldridge 2010; Sun and Shapiro 2022). We further include county-by-rural fixed effects to capture time-invariant rural–urban differences within counties, such as baseline gaps in school quality or local labor-market opportunities. Identification comes from contrasts between rural and urban individuals born in the same county and cohort; under the assumption that, absent the reform, these within-county rural–urban gaps would have followed parallel trends across cohorts, the resulting estimates can be given a causal interpretation.

In addition to the cohort-based difference-in-differences design, we also exploit geography-based variation in county-level pre-reform academic high school capacity. Specifically, we proxy pre-policy capacity using cohort-level academic high school completion rates for pre-reform cohorts and examine how post-reform rural–urban convergence in academic high school completion varies across counties with pre-reform capacity. At the province level, we further examine heterogeneity by provincial higher-education expansion intensity and by the density of four-year universities.

Our findings indicate that the expansion significantly increased rural students' like-

lihood of completing the academic high school track. Difference-in-differences estimates with county and cohort fixed effects show that, following the reform, the rural–urban gap in academic high school completion narrowed by 2.9 percentage points, relative to a pre-expansion rural completion rate of only 16.7 percent. The rural–urban gap in average years of schooling also narrowed by 0.26 years, consistent with the policy having raised perceived returns to continued schooling and encouraged rural students to remain in school longer. In estimates separately by gender, the increase in academic high school completion is substantially larger among rural males than among rural females, whereas the gains in years of schooling, though still somewhat larger for males, are much closer in magnitude across genders (Goldin, Katz, and Kuziemko 2006; Heath and Jayachandran 2017).

Beyond these main estimates, we document an expectations-based response that is strongly shaped by local constraints. The geography-based exposure analysis shows that rural gains are much larger in counties with greater pre-policy academic high school capacity, where the academic track was effectively more attainable prior to the reform. Cohorts in higher-capacity counties exhibit markedly stronger rural–urban convergence, and the estimated post-reform rural gain relative to urban increases by about 1.7 percentage points for each one–standard-deviation rise in pre-reform academic high school capacity. This pattern is consistent with higher perceived attainability of the academic track among marginal high-ability rural students, even in settings where rural–urban gaps in tertiary completion persist (Li et al. 2015). At the provincial level, rural schooling gains are also stronger in provinces that expanded university places more aggressively and in those with denser four-year university systems. These patterns suggest that both the tightness of upper-secondary capacity constraints and the scale and structure of the tertiary system shape how strongly rural students update beliefs and adjust upper-secondary track choices in response to the expansion.

The upward shift in educational trajectories is accompanied by an improvement in occupational outcomes. The share of rural respondents employed in low-skilled sectors such as agriculture, manufacturing, or construction declined by 5.1 percentage points relative to their urban counterparts, while the probabilities of working in high-skill industries and in jobs requiring an academic high school education increased by 3.2 and 4.5 percentage points, respectively. This reallocation toward higher-skill jobs is consistent with evidence that concentrations of educated workers generate productivity and employment spillovers in urban and non-farm sectors (Glaeser and Maré 2001; Rauch 1993; Charlot and Duranton 2004; Mazzolari and Ragusa 2013). Several robustness checks support the validity of our findings. A battery of falsification exercises and alternative sample definitions yields no evidence of spurious pre-trends or cohort-specific artifacts in rural–urban educational or occupational gaps, bolstering a causal interpretation of our results (Huang, Lei, and

Sun 2021; Goodman-Bacon 2021).

This paper makes three contributions. First, we report internally-valid and precise estimates of a major nationwide higher-education policy on upstream upper-secondary tracking decisions. Relative to studies that focus on college enrollment or income (Che and Zhang 2018; Belskaya, Peter, and Posso 2020; Blundell, Green, and Jin 2022; Mbiti et al. 2019), we shift attention to the academic-versus-vocational margin and provide comprehensive evidence on how the 1999 expansion changed academic high school participation, especially for rural students. We show that the reform increased academic-track take-up among rural students, narrowing the rural–urban gap in academic high school completion even as some outside evidence points to widening rural–urban differences in college completion.

Second, we show that the upstream impacts of a nationwide tertiary expansion are highly place-dependent. The reform's effects on academic high school participation and years of schooling are substantially larger in counties with greater pre-reform academic high school capacity and in provinces that expanded university places more aggressively, highlighting how local constraints and place-based resources mediate the distributional consequences of national education reforms (Moretti 2004; Chetty et al. 2022).

Third, we link these schooling responses to occupational sorting. Rural cohorts more exposed to the reform are less likely to enter low-skilled sectors, and are correspondingly more likely to work in high-skill industries and in jobs that typically require an academic high school education. This suggests that improved access to the academic track not only lengthened schooling but also reallocated workers toward higher-skill placements (Autor and Handel 2013; Dominguez and Ruffini 2023).

The remainder of the paper is structured as follows. Section 2 provides institutional background on China's education system and the 1999 higher-education expansion. Section 3 describes the data sources, variable construction, and descriptive statistics. Section 4 outlines the empirical strategy. Section 5 presents the main results on educational attainment and labor-market outcomes. Section 6 conducts a series of robustness checks, and Section 7 addresses potential confounding factors. Section 8 explores heterogeneous effects across regions and subgroups. Finally, Section 9 concludes with policy implications and directions for future research.

# 2 Institutional Background

This section provides institutional background on China's pre-tertiary schooling system and the 1999 expansion of higher education, with particular emphasis on the upper-secondary tracking margin that underpins our empirical design. After completing nine years of compulsory education, students must choose between academic high school (the

academic track) and secondary vocational school (the vocational track). Admission to university is almost entirely contingent on passing the National College Entrance Examination (*Gaokao*), which in practice is restricted to graduates of academic high schools. This institutional structure creates a sharp tracking margin at the transition to upper secondary: students who enter the vocational track are effectively excluded from regular tertiary channels.

Before 1999, extremely limited university capacity made this academic path especially unattractive for rural students. The raw national college admission rate was below 10%, and the combined direct and opportunity costs of three years of academic high school often exceeded the annual income of a rural household. In contrast, vocational schools charged lower tuition, were eligible for local subsidies, and offered a more direct pathway into the job market. Given these constraints, many rural families preferred the vocational track even when some of their children had academic ability comparable to, or exceeding, that of their urban counterparts.

In June 1999, the central government approved a major expansion of higher education. That year, the number of university admissions rose by more than 47% relative to the previous year, initiating a sustained expansion trajectory. Because the decision was made only after the completion of the 1998 university admission cycle, students who had already entered upper secondary school could not revise their educational paths in response. By contrast, students who were still in junior secondary school when the expansion was announced could reassess their upper-secondary track choice in light of higher perceived returns to tertiary study. The reform therefore generated a sharp, nationwide increase in the expected payoff to pursuing the academic track for cohorts still at the upper-secondary margin.

Our empirical analysis treats cohorts born in 1983–1988 as reform-exposed and those born in 1976–1982 as pre-reform. This classification is driven by the timing of schooling relative to the 1999 announcement. Under China's standard schooling calendar, students typically enter primary school at age seven and progress through the 6–3–3 structure. The 1983 birth cohort is thus the first for which the June 1999 expansion could plausibly affect the choice between the academic and vocational upper-secondary tracks. Table A1 summarizes the mapping from birth cohort to schooling stage in the 1998–1999 school year, showing that cohorts born in 1981–1982 were already in upper secondary when the reform was announced, while cohorts born in 1976–1980 had completed upper secondary or left school entirely; cohorts born in 1983 and later were still in junior secondary at the time of track choice. Deviations from the modal pattern due to early or late school entry or grade repetition blur this boundary somewhat, but primarily generate small amounts of classical misclassification that attenuate differences between adjacent cohorts, making our estimates conservative.

Despite the national scope of the reform, China's education markets remained segmented by enduring urban–rural divides. Rural students continued to face disadvantages in school quality, exam-preparation resources, and family support, all of which made it more difficult for them to translate the expanded college places into actual admission opportunities. These frictions imply that the 1999 policy did not generate uniform gains at the tertiary tier and that earlier margins in the schooling sequence are crucial for understanding its distributional consequences. For expositional purposes, it is useful to view the education sequence as involving two linked stages. At the first stage—the choice between academic-track and vocational-track high school—the nationwide relaxation of university admission thresholds raised the expected payoff to the academic track, especially for students near the relevant cutoff and in counties where pre-policy upper-secondary capacity was greater and admission competition less intense. At the second stage—college admission—continued selectivity and path dependence (differences in feeder-school quality, counseling, and the private costs of exam retakes) limited full convergence. This motivates our focus on behavioral responses at the upper-secondary transition; the subsequent sections document how these responses translate into changes in schooling attainment and a reallocation away from low-skilled employment.

## 2.1 China's Education System

China's pre-tertiary system follows a 6–3–3 structure: six years of primary school, three years of junior secondary, and three years of upper secondary. Nine-year compulsory education covers primary and junior secondary grades (roughly ages 6 to 15). Upon completing compulsory schooling, students face a critical juncture: they either enroll in an academic high school or in a secondary vocational school, a choice that largely determines their subsequent opportunities for tertiary education.

Academic high schools offer three years of general academic coursework that prepares students for the National College Entrance Examination, China's nationwide college entrance exam. Admission to universities at either the bachelor's (four-year) or associate (three-year) level is almost entirely contingent on performance on this exam, and the overwhelming majority of candidates come from the academic track. Vocational high schools also span three years but focus on technical and skill-based training, leading primarily to direct employment upon graduation. A small number of specialized "bridging" programs allow some vocational graduates to pursue higher education, but the mainstream university pathway runs overwhelmingly through the academic track.

Figure 1 illustrates the structure of the upper-secondary education system in China and shows how these two tracks connect to subsequent educational and occupational opportunities.

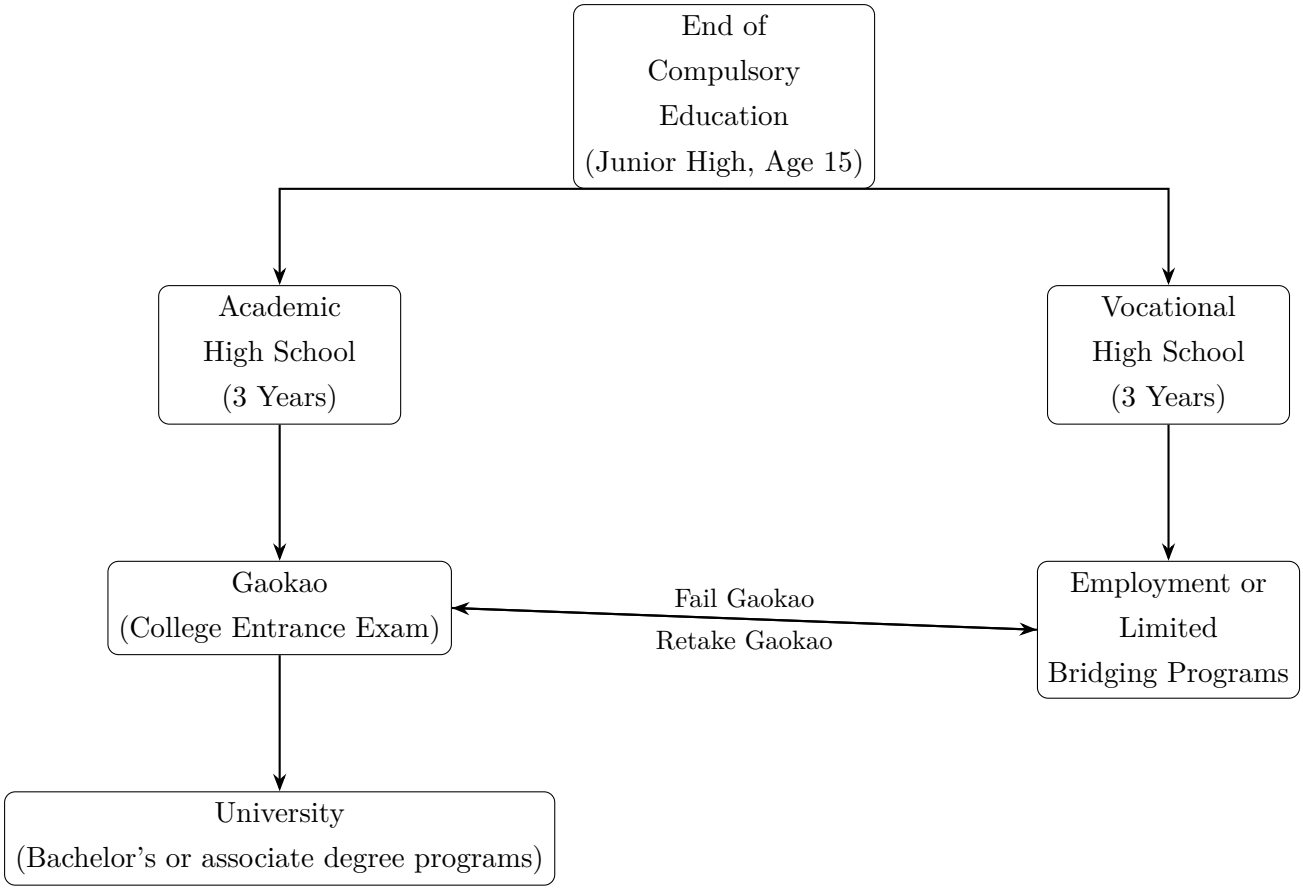


Figure 1: Overview of the Chinese Upper-Secondary Education Tracks

## 2.2 Upper-Secondary Admissions and Capacity

Most students, whether they enroll in academic or vocational high schools, attend institutions located within their county of registration. In China, both types of high schools are typically administered by county-level education authorities, which are responsible for their funding and daily operation.[3] Placement into academic high schools and secondary vocational schools is determined by a unified entrance examination with centralized assignment at the county level. During our period, admission to academic high school was competitive and rationed. Long-standing policy guidance to maintain a rough balance between academic and vocational intakes limited the scope for rapid, unilateral expansions of academic high school seats.

Central guidance in the mid-1990s framed academic high schooling as one component of a balanced upper-secondary system rather than a sector for aggressive expansion. A 1995 circular on running academic high schools instructed that large and medium cities and economically more developed rural areas should expand academic high schools gradually and in proportion to the need to universalize upper-secondary education, that

[3]See the *Education Law of the People's Republic of China* and Ministry of Education yearbooks for the governance of pre-tertiary education at the county level.

counties with intermediate development should expand only modestly in light of local conditions, and that poorer counties should first complete nine-year compulsory schooling while continuing to adjust the structure of secondary education.[4]

A related supply-side feature of this period is the nationwide push to consolidate rural school networks in the early 2000s, which generally encouraged the closure or merger of small, dispersed teaching sites and village schools and a shift toward more centralized provision in township or county-seat schools. Although this policy primarily targeted compulsory education, it provides useful institutional context for our setting: it points to an environment of consolidation and concentration in rural schooling infrastructure rather than a broad-based expansion in physical access. We therefore view the post-1999 rural gains documented below as unlikely to be mechanically driven by uniformly easier geographic access to academic high schools. As contextual evidence, Appendix Table A2 summarizes changes in rural school counts over time, which are broadly consistent with consolidation. We emphasize that these patterns are presented as background rather than as a source of identifying variation.

The 1999 university expansion directly targeted the tertiary tier. Upper-secondary capacity adjusted only gradually and unevenly across space: some jurisdictions added classes, upgraded selected schools, or opened new campuses, but changes were modest relative to the tertiary surge and tilted toward urban, academically selective schools. Vocational capacity also expanded to preserve the academic–vocational balance. Consistent with gradual adjustment at the upper-secondary level, national admissions series show a steady multi-year rise in senior high admissions over 1995–2003 with no discrete jump in 1999, whereas university admissions were nearly flat in 1995–1998 and surged from 1999 onward (Appendix Figure A4).[5]

Rural applicants faced the same formal capacity constraints but higher effective costs (e.g., boarding and commuting to county-seat schools) and weaker preparation in feeder junior highs. Many rural counties offered fewer academic high school seats per cohort and imposed higher cutoffs than urban districts, placing rural students closer to the academic–vocational margin.

In this environment, the 1999 reform operated primarily through expectations at the upper-secondary transition: the surge in university places raised the perceived payoff to choosing the academic track for students who had not yet been tracked. A standard

[4]State Education Commission, "Several Opinions on Running General Senior High Schools," 1995. Policy wording (authors' translation): "Large and medium-sized cities and economically more developed rural areas should develop academic high schools in a planned and proportional manner according to the needs of universalizing upper-secondary education; regions with intermediate levels of economic development should expand only to a moderate extent based on local realities; economically less developed regions should first accomplish the task of universalizing nine-year compulsory education while continuing to adjust the structure of secondary education."

[5]Admissions series compiled from *China Statistical Yearbook*, various years; authors' calculations. We use these series as supply-side background rather than for identification.

educational pipeline implies that behavioral responses should be strongest where local constraints are least binding and where access to tertiary study is most salient. Counties with greater pre-policy academic high school capacity offer rural students more scope to translate higher expected returns into academic high school enrollment, whereas in capacity-constrained counties tighter admission bottlenecks limit such adjustments. At the tertiary tier, provinces with denser university systems and more aggressive post-1999 expansion make the college pathway more visible and credible, potentially amplifying these responses. These mechanism-based predictions motivate the heterogeneity and triple-difference analyses below. Formal identification and the difference-in-differences specification are presented in Section 4; here we focus on how these institutional features shape the interpretation of our estimates.

# 3 Data

## 3.1 2015 1% Population Sample Survey of China

We use microdata from the 2015 1% Population Sample Survey of China as our primary data source, a nationally representative microdataset administered by the National Bureau of Statistics. The survey is drawn from the full decennial census and covers roughly one percent of the national population, comprising over one million individuals. It records detailed demographic, educational, occupational, and geographic information, making it one of the most authoritative sources for studying long-run socioeconomic outcomes at the individual level. Crucially, the dataset includes information on parental education, employment industry, and land-contract arrangements, which we use to construct key variables on treatment exposure, rural versus urban status, and heterogeneous effects.

The 2015 wave provides a unique opportunity to assess the consequences of education reforms from the late 1990s. It observes completed educational attainment and early labor-market outcomes for individuals born in the 1980s, a cohort directly affected by the 1999 expansion of higher education. Crucially, it is the only census-scale dataset that reports full schooling outcomes while distinguishing academic from vocational programs at the upper-secondary level.

We restrict our sample to individuals born between 1976 and 1988, a 13-cohort window that spans the pre- and post-reform births relevant for our design. Cohorts at the upper end of this window (1976–1982) were already enrolled in upper-secondary school when the June 1999 university expansion was announced and serve as an unaffected baseline, while cohorts at the lower end (1983–1988) were still in lower secondary school or younger and had full scope to revise their educational plans.[6] After dropping observations with

[6]Including earlier birth cohorts would add many observations but little additional identifying variation, as their schooling and initial labor-market decisions were completed well before the 1999 expansion

missing covariates, the working sample comprises 274,576 individuals.

## 3.2 Key variables

Our analysis tracks education along two complementary dimensions. We first construct binary indicators for completing each credential in the education system, namely academic high school, vocational high school, junior college, and university, to characterize shifts in the distribution of qualifications. We then construct a continuous measure of accumulated years of schooling to capture intensive-margin responses. Together, these variables provide a comprehensive view of the extensive and intensive effects of the 1999 university expansion.

For labor-market status, the census does not report income, so we proxy economic position using the respondent's main occupation (national three-digit occupation codes). We aggregate occupations into broad groups and define a baseline indicator for low-skilled work, covering agricultural, basic service, and production/transport jobs; other occupations form the comparison category. In some specifications we also distinguish high-skill managerial and professional occupations, as well as occupations that typically require at least upper-secondary schooling. Although occupation codes do not reveal within-occupation wage dispersion, they summarize typical pay, required skills, and working conditions and are a standard proxy for economic position when earnings are unavailable. Appendix Table A3 documents the mapping from the original occupation codes to our ten-category aggregation.

We also construct geography-based exposure measures that enter the main empirical design. To capture pre-policy academic high school capacity at the upper-secondary margin, we build a county-level index using academic high school completion among pre-policy cohorts (1976–1982): for each county, we average the share completing the academic track across these cohorts and standardize this average across counties. In the institutional context of China's tracked secondary system, academic high school completion is tightly linked to seat availability and therefore provides a practical proxy for local academic high-school capacity at the county level. Higher values of this index indicate counties where the academic track was easier to access before the reform. In these counties, competition for academic high school places was less intense, and rural students faced lower effective cutoffs than in more constrained counties. Because it is based entirely on pre-policy cohorts, this index is not mechanically affected by the reform and can be viewed as a proxy for pre-1999 academic high-school capacity. As a downstream exposure, we proxy tertiary proximity with province-level university density measured

and under different institutional conditions. Extending the window to cohorts born after 1988 would both compress the time available to observe stable labor-market outcomes by 2015 and increase potential contamination from subsequent education reforms, notably the 2006–2007 abolition of rural compulsory-education fees in rural areas.

before 1999. Controls include gender, ethnicity, and the highest parental education.

Because the census does not record hukou status directly, we proxy rural registration using land-contract rights: we classify a household as rural if it holds a long-term contract to collectively owned village farmland, and as urban otherwise. Under China's land system, eligibility for long-term contracts to collectively owned village farmland is institutionally tied to rural registration, so land-contract rights serve as a tight proxy for rural registration.

## 3.3 Descriptive statistics

Table 1 summarizes means for key outcomes by cohort group and rural–urban status, contrasting pre-expansion (1976–1982) and post-expansion (1983–1988) cohorts. Before the policy, rural youth averaged 9.4 years of schooling and only 16.7 percent completed the academic track (academic high school plus college or higher); the corresponding urban figures were 12.3 years and 55.1 percent. After the reform, the rural–urban gap narrows markedly in both metrics. Consistent with these schooling gains, the share of rural workers in low-skilled occupations declines more sharply than the urban share. These patterns foreshadow the difference-in-differences estimates presented below.

The census microdata thus provide a convenient platform for tracing how a large nationwide relaxation of university capacity propagated back through the secondary-school pipeline and into the labor market. The auxiliary yearbook series, in turn, allow us to condition on local trends in economic structure and educational resources and to explore heterogeneous responses across different county and provincial environments.

Table 1: Descriptive Statistics Before and After the 1999 Expansion

| | Pooled | | Urban | | Rural | |
|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD |
| *Panel A: Pre-Expansion Cohorts (1976–1982)* | | | | | | |
| *Controls* | | | | | | |
| Age | 35.40 | 1.72 | 35.37 | 1.73 | 35.42 | 1.72 |
| Male | 0.486 | 0.50 | 0.538 | 0.50 | 0.453 | 0.50 |
| Ethnic minority | 0.094 | 0.29 | 0.075 | 0.26 | 0.107 | 0.31 |
| Father's education | 8.46 | 1.30 | 8.76 | 1.35 | 8.26 | 1.22 |
| Mother's education | 6.98 | 1.74 | 7.39 | 1.81 | 6.69 | 1.64 |
| *Outcomes* | | | | | | |
| Years of schooling | 10.59 | 2.98 | 12.29 | 3.21 | 9.40 | 2.11 |
| Academic high school | 0.135 | 0.34 | 0.551 | 0.36 | 0.167 | 0.32 |
| Vocational high school | 0.068 | 0.25 | 0.105 | 0.31 | 0.042 | 0.20 |
| College or higher | 0.189 | 0.39 | 0.393 | 0.49 | 0.047 | 0.21 |
| Low-skill occupation | 0.526 | 0.50 | 0.377 | 0.48 | 0.630 | 0.48 |
| *N* | 138,156 | | 55,899 | | 82,257 | |
| *Panel B: Post-Expansion Cohorts (1983–1988)* | | | | | | |
| *Controls* | | | | | | |
| Age | 29.31 | 1.70 | 29.34 | 1.70 | 29.28 | 1.69 |
| Male | 0.494 | 0.50 | 0.542 | 0.50 | 0.462 | 0.50 |
| Ethnic minority | 0.096 | 0.29 | 0.090 | 0.29 | 0.100 | 0.30 |
| Father's education | 9.13 | 1.19 | 9.33 | 1.29 | 8.99 | 1.10 |
| Mother's education | 7.94 | 1.58 | 8.20 | 1.71 | 7.75 | 1.47 |
| *Outcomes* | | | | | | |
| Years of schooling | 10.99 | 3.02 | 12.40 | 3.21 | 10.03 | 2.46 |
| General high school | 0.383 | 0.36 | 0.573 | 0.36 | 0.254 | 0.35 |
| Vocational high school | 0.065 | 0.25 | 0.083 | 0.28 | 0.053 | 0.22 |
| College or higher | 0.234 | 0.42 | 0.422 | 0.49 | 0.107 | 0.31 |
| Low-skill occupation | 0.497 | 0.50 | 0.371 | 0.48 | 0.582 | 0.49 |
| *N* | 137,016 | | 55,391 | | 81,625 | |

*Notes::* Based on the 2015 1% Population Sample Survey of China. Schooling variables follow standard definitions; parental education is measured in years. Panel A reports cohorts born 1976–1982 (pre-expansion) and Panel B reports cohorts born 1983–1988 (post-expansion).

# 4 Empirical Strategy

## 4.1 Baseline DID specification

We begin by estimating a difference-in-differences model that compares rural and urban individuals born in different cohorts within the same county. Let $Y_{icy}$ denote the outcome for individual $i$ residing in county $c$ and born in cohort $y$. Define $\text{Rural}_{ic} \in \{0, 1\}$ as an indicator for rural registration, and let $\text{Post}_y = \mathbb{1}\{y \geq 1983\}$ denote post-reform cohorts. Let $\mathbf{X}_{icy}$ collect individual controls (gender, minority status, and parental education). Our baseline difference-in-differences specification is

$$Y_{icy} = \alpha + \beta \left(\text{Rural}_{ic} \times \text{Post}_y\right) + \mathbf{X}'_{icy}\gamma + \mu_c + \lambda_y + \eta_{c\times\text{Rural}} + \varepsilon_{icy}, \tag{1}$$

where $\mu_c$ are county fixed effects that absorb time-invariant cross-county differences, $\lambda_y$ are cohort fixed effects that capture cohort-wide shocks common to all individuals born in year $y$, and $\eta_{c\times\text{Rural}}$ are county-by-rural fixed effects that absorb county-specific level differences. Standard errors are clustered at the county level throughout. In (1), the coefficient $\beta$ captures how the rural–urban gap in $Y$ changes between pre- and post-reform cohorts within counties, after controlling for covariates and the full set of fixed effects. Lower-order main effects of $\text{Rural}_{ic}$ and $\text{Post}_y$ are collinear with the absorbed fixed effects and are therefore not separately identified.

In China's pre-tertiary system, funding, staffing, and oversight of junior and upper-secondary schools are predominantly organized at the county level. Our design compares rural and urban individuals born in the same county and cohort under a common county budget and policy environment. County fixed effects absorb time-invariant cross-county differences in school capacity, financing, and economic structure; cohort fixed effects absorb nationwide shocks common to each birth cohort; and county-by-rural intercepts allow for persistent within-county rural–urban level differences. Identification of $\beta$ comes from differential changes in outcomes between rural and urban individuals across pre- and post-reform cohorts within counties. Under the maintained assumption that, absent the reform, these within-county rural–urban gaps would have followed parallel trends across cohorts, $\beta$ has a causal interpretation.

## 4.2 Event-Study Specification

To examine cohort-by-cohort dynamics and assess pre-trends, we estimate an event-study version of the baseline specification. As above, individuals born in 1983 or later are classified as post-expansion cohorts: these cohorts were still in lower-secondary school when the 1999 university expansion was announced and could adjust upper-secondary track choices in response to the policy. By contrast, those born before 1983 had already

entered high school and were effectively unaffected by the reform.

To visualize dynamic effects, we estimate:

$$Y_{icy} = \sum_{y' \neq 1982} \beta_{y'} \, \mathbb{1}\{\text{Birthyear}_i = y'\} \times \text{Rural}_{ic} + \mathbf{X}'_{icy}\gamma + \mu_c + \lambda_y + \eta_{c\times\text{Rural}} + \varepsilon_{icy}, \quad (2)$$

where $\mu_c$ are county fixed effects that absorb time-invariant county characteristics, $\lambda_y$ are cohort fixed effects that capture shocks common to all individuals born in year $y$, and $\eta_{c\times\text{Rural}}$ are county-by-rural fixed effects that absorb county-specific level differences. Standard errors are clustered at the county level. The coefficients $\beta_{y'}$ trace, cohort by cohort, how the within-county difference in $Y$ between rural and urban individuals evolves relative to the 1982 cohort, after controlling for covariates and fixed effects. We normalize $\beta_{1982} = 0$ so that all coefficients are interpreted relative to the 1982 cohort. Flat and statistically insignificant pre-reform coefficients $\beta_{y'}$ for $y' < 1983$ are consistent with parallel trends in rural–urban gaps prior to the reform.

To complement the event-study, Figure A2a plots raw cohort-level means for the treatment (rural) and control (urban) groups. The unadjusted series exhibit approximately parallel levels and trends pre-reform, followed by a gradual divergence for post-reform cohorts.

## 4.3 Geography-based exposure: construction and design

In addition to the cohort-based rural–urban difference-in-differences in (1), we incorporate the geography-based exposure measures defined in Section 3. The key object is a county-level index of pre-policy academic high school capacity at the upper-secondary margin. For each county, we average academic high school completion rates across pre-reform cohorts (1976–1982) and standardize this average across counties. As discussed in Section 3, higher values of this index indicate counties where the academic track was easier to access before the reform: competition for academic high school places was less intense, and rural students faced lower effective cutoffs than in more constrained counties. Because the index is constructed entirely from pre-policy cohorts, it is not mechanically affected by the 1999 expansion and can be interpreted as a proxy for the capacity of the academic high school seats.

This geography-based dimension provides a second source of identifying variation. The baseline specification in (1) compares rural–urban gaps across pre- and post-reform cohorts within counties; the geography-based design asks whether this rural–post contrast is systematically larger in counties where the academic track was ex ante easier for rural students to access. To implement this design, we harmonize county identifiers across census sources and consolidate any post-baseline splits or mergers back to the baseline county boundary, with individuals $i$ nested in counties $c$ and cohorts $y$ denoting birth

years.

Formally, we augment the baseline DID equation with a triple interaction between rural status, post-reform cohort, and county-level capacity:

$$\begin{aligned} Y_{icy} = \alpha + \beta\,(\text{Rural}_{ic} \times \text{Post}_y) + \theta\,(\text{Rural}_{ic} \times \text{Post}_y \times \text{Capacity}_c) + \mathbf{X}'_{icy}\gamma \\ + \mu_c + \lambda_y + \eta_{c\times\text{Rural}} + \varepsilon_{icy}. \end{aligned} \tag{3}$$

Here, $\mu_c$ are county fixed effects that absorb time-invariant cross-county differences, $\lambda_y$ are cohort fixed effects that capture shocks common to all individuals born in year $y$, and $\eta_{c\times\text{Rural}}$ are county-by-rural fixed effects that absorb county-specific level differences. Because $\text{Capacity}_c$ is constant within a county, its main effect is absorbed by $\mu_c$, and its interaction with rural status is absorbed by $\eta_{c\times\text{Rural}}$; we omit these lower-order terms without loss of generality. Equation (3) thus isolates how the within-county rural–post contrast covaries with pre-policy capacity. The coefficient $\theta$ captures whether the reform-induced narrowing of rural–urban gaps is larger in counties where the academic track was easier to access before the reform. Standard errors are clustered at the county level. Identification requires that, conditional on fixed effects and covariates, the counterfactual evolution of rural–urban differences within a county would have been parallel across values of $\text{Capacity}_c$ in the absence of the reform.

# 5 Results

## 5.1 Effects on Years of Schooling

Table 2 revisits prior evidence that the 1999 expansion raised average schooling for rural cohorts relative to urban cohorts. Using improved data and a design that aligns the analysis with the junior-high transition, we recover a clear pattern: pre-reform cohorts show no systematic rural–urban differences, consistent with flat pre-trends, while the gap narrows sharply beginning with the 1983 cohort and continues to close through 1988. The average post-reform effect is sizable in the pooled sample and is noticeably larger for men than for women. Cohort-specific estimates are economically meaningful—for example, the rural–urban differential declines by roughly 0.19 years for the 1985 cohort and by about 0.46 years for the 1988 cohort. These results confirm, on better data, that even if gaps at the tertiary margin may have widened, gains accumulated at earlier stages of schooling produced marked convergence in total years of schooling, particularly among rural men.

Table 2: Effect of the 1999 Expansion on Years of Schooling

| | **Years of Schooling** | | |
|---|---|---|---|
| | **Pooled** | **Male** | **Female** |
| Rural × Post | 0.255*** | 0.298*** | 0.226*** |
| | (0.024) | (0.031) | (0.029) |
| Controls | Yes | Yes | Yes |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Observations | 274,576 | 139,873 | 134,431 |
| $R^2$ | 0.370 | 0.377 | 0.393 |
| Clustering Level | County | County | County |

*Notes*: The dependent variable is completed years of schooling. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include sex (pooled sample), ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * p < 0.10, ** p < 0.05, *** p < 0.01.

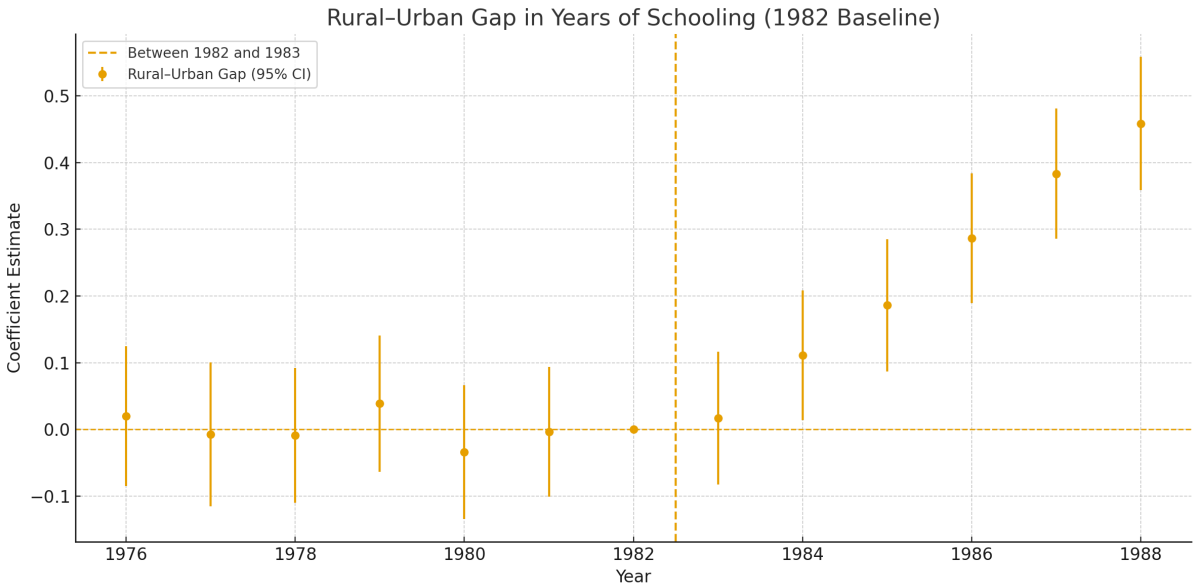


Figure 2: Rural–urban gaps in years of schooling by birth cohort

*Notes*: Points plot estimated coefficients $\hat{\beta}_k$ from the event-study specification in equation (2), where $k$ indexes birth cohorts relative to the omitted 1982 cohort ($k = 0$). The outcome is the rural–urban difference in completed years of schooling. Regressions include county, birth-cohort, and county-by-rural fixed effects and control for gender, ethnicity, and parental education. Vertical bars show 95% confidence intervals based on standard errors clustered at the county level.

## 5.2 Main result: Expansion Raised Rural Academic High School Completion

Table 3 shows that the 1999 university-expansion policy significantly increased the likelihood that rural youth completed the academic high school track. In the pooled sample, the rural–cohort interaction coefficient is 0.029, implying a 2.9 percentage points increase in academic high school completion for post-expansion rural cohorts relative to their urban counterparts. Given a baseline completion rate of approximately 16.7 percent among rural students, this represents an effect size of about 17 percent relative to the pre-expansion mean and signals meaningful progress toward narrowing rural–urban disparities in upper-secondary education.

Estimating the regression separately by gender yields positive and statistically significant effects for both boys and girls, but with a larger magnitude for boys (around 4 percentage points) than for girls (around 2 percentage points). These patterns indicate that the reform expanded academic high school access for rural youth of both sexes, while rural boys experienced greater relative gains. To connect these results to the university margin and to the existing literature, Appendix Table A5 reports analogous specifications for higher-education completion. Consistent with prior work on the 1999 expansion, the rural–urban gap in higher-education attainment widens slightly after the reform, and this shift displays the same gender asymmetry documented above.

Several structural factors may contribute to the persistence of gender gaps in overall educational attainment. First, rural areas generally suffer from weaker educational infrastructure, including lower-quality schools, limited access to qualified teachers, and fewer pathways to post-secondary education. When educational opportunities are scarce, families may prioritize sons over daughters due to entrenched gender norms that value male education more highly. In relatively disadvantaged regions where traditional attitudes remain strong, the preference for educating boys is often amplified by limited household resources. Additionally, the perceived higher economic returns to male education in the labor market may further reinforce this bias. As a result, even as aggregate access expands, underlying gender asymmetries in educational investment may persist, particularly in low-income or less-developed rural areas.

Finally, cohort-specific estimates from the event-study specification indicate that the earliest post-reform coefficients are imprecisely estimated and not statistically distinguishable from zero, but they gradually move from slightly negative to slightly positive values across exposed cohorts. This timing is consistent with several features of the reform environment. First, despite high year-on-year growth rates in university admissions immediately after 1999, the absolute increase in places was initially modest relative to the size of upper-secondary cohorts. Second, on the supply side of upper-secondary education, adjustments in academic high school capacity, staffing, and tracking thresholds

plausibly occurred with delays and in discrete steps. Third, information about the increased expected returns to the academic track likely diffused gradually, particularly in rural areas. As perceived admission probabilities rose and search and application frictions diminished, rural students appear to have increasingly selected into the academic track, yielding progressively larger rural gains in academic high school completion for later cohorts. A similar delayed response pattern is documented for targeted education transfers in Chile (Lucas, McEwan, and Torres Irribarra 2025).

Table 3: Effect of the 1999 Expansion on Academic High School Completion

| | **Completed Academic High School** | | |
|---|---|---|---|
| | **Pooled** | **Male** | **Female** |
| Rural × Post | 0.029*** | 0.040*** | 0.020*** |
| | (0.004) | (0.005) | (0.005) |
| Controls | Yes | Yes | Yes |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Observations | 274,576 | 139,873 | 134,431 |
| $R^2$ | 0.257 | 0.262 | 0.285 |
| Clustering Level | County | County | County |

*Notes*: The dependent variable equals 1 if an individual completed an academic high-school track. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include sex (pooled sample), ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

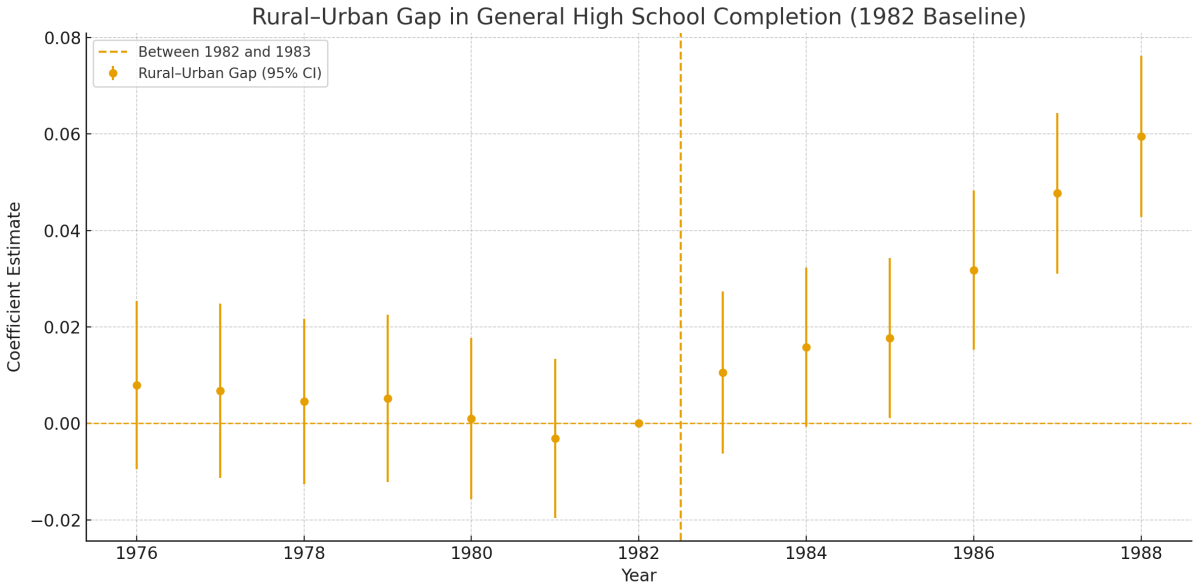


Figure 3: Academic High School

*Notes*: Points plot estimated coefficients $\hat{\beta}_k$ from the event-study specification in equation (2), where $k$ indexes birth cohorts relative to the omitted 1982 cohort ($k = 0$). The outcome is the rural–urban difference in academic high school completion. Regressions include county, birth-cohort, and county-by-rural fixed effects and control for gender, ethnicity, and parental education. Vertical bars show 95% confidence intervals based on standard errors clustered at the county level.

## 5.3 Main Result: How Did the Expansion Affect Long-Term Occupational Outcomes?

Table 4 summarizes how the 1999 expansion reshaped the long-run occupational structure of rural cohorts. Column (1) shows a sizable shift out of low-skill occupations for rural cohorts exposed to the reform, relative to urban cohorts from the same counties and birth cohorts. The estimated coefficient on the Rural×Post term is $-0.049$, implying that the expansion reduced the probability that a rural individual works in an agricultural or production/transport occupation by about five percentage points relative to the counterfactual path. Given the high pre-reform incidence of such jobs among rural youth, this represents an economically meaningful decline in low-skill employment.

Columns (2) and (3) show that these losses in low-skill employment are accompanied by gains in higher-quality jobs. The Rural×Post interaction increases the probability of working in a high-skill occupation by 3.2 percentage points and the probability of holding an occupation that typically requires at least upper-secondary schooling by 4.5 percentage points. In other words, exposed rural cohorts are not simply moving from one type of low-skill work to another; they are more likely to end up in white-collar or otherwise skill-intensive positions whose entry margins are tightly linked to schooling.

A natural concern is that a nontrivial share of individuals are not employed at the time of the census, so occupational outcomes are observed only for a selected subset of the population. Our main specifications address this directly in the construction of the outcome variables. Using the survey question on the main reason for not working, we first distinguish students, retirees, those who have lost the ability to work, and individuals engaged in full-time home production from those with unemployment-like reasons (graduates who have never worked, job loss for firm or personal reasons, or land expropriation). The former groups are excluded from the occupational analysis because they are not actively in the labor market. Individuals with unemployment-like reasons are instead coded as having an unfavorable labor-market outcome and are folded into the low-skill category in our main measure. Thus, the estimated Rural×Post effects capture both movements up the job-quality ladder among the employed and changes at the extensive margin between employment and unemployment.

Taken together with the earlier results on academic high school completion and years of schooling, these patterns indicate that expanding university opportunities did more than raise educational attainment on paper. The reform durably reallocated rural youth up the occupational ladder—away from low-skill manual work and into occupations that make more intensive use of the additional human capital generated at the upper-secondary and tertiary margins.

Table 4: Effect of the 1999 Expansion on Occupational Sorting

| | **Labor Market Outcomes** | | |
|---|---|---|---|
| | **Low-skill occ.** | **High-skill occ.** | **HS-required occ.** |
| Rural × Post | -0.049*** | 0.032*** | 0.045*** |
| | (0.004) | (0.003) | (0.004) |
| Controls | Yes | Yes | Yes |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Observations | 254,597 | 254,597 | 254,597 |
| $R^2$ | 0.248 | 0.136 | 0.194 |
| Clustering Level | County | County | County |

*Notes*: Each column reports estimates from a separate high-dimensional fixed-effects regression. The dependent variable is an indicator for employment in a low-skill occupation (column 1), a high-skill occupation (column 2), or an occupation that typically requires at least upper-secondary schooling (column 3). All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include sex, ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

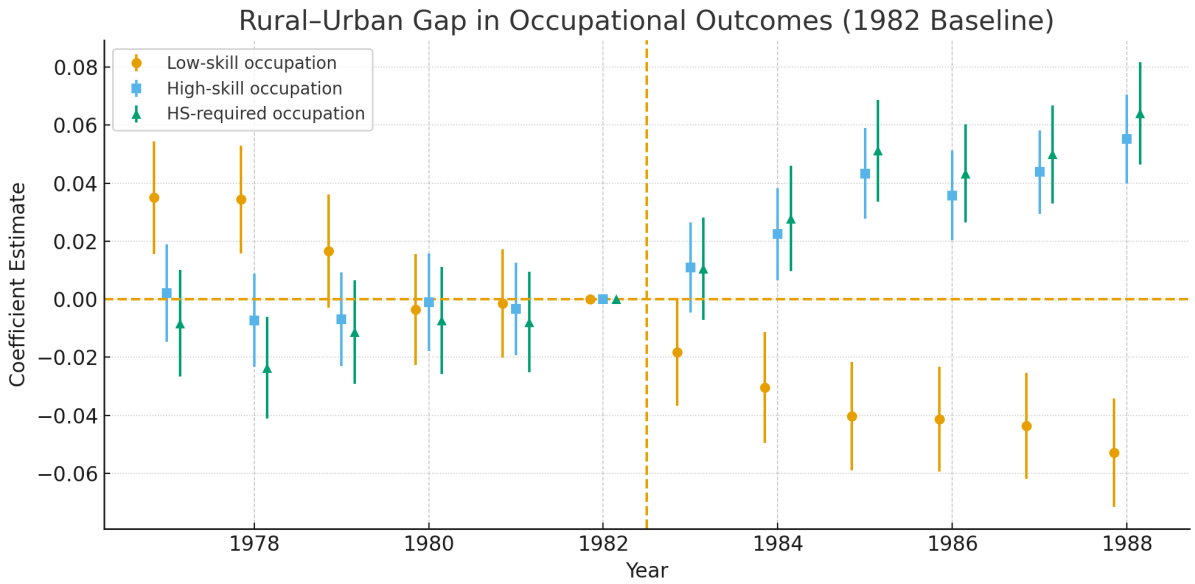


Figure 4: Rural-urban gaps in different industries

*Notes*: Points plot estimated coefficients $\hat{\beta}_k$ from the event-study specification in equation (2), where $k$ indexes birth cohorts relative to the omitted 1982 cohort ($k = 0$). The outcome is the rural–urban difference in probability of working in different types of occupation. Regressions include county, birth-cohort, and county-by-rural fixed effects and control for gender, ethnicity, and parental education. Vertical bars show 95% confidence intervals based on standard errors clustered at the county level.

**Economic Significance of the Main Effects** The expansion raised rural academic high school completion by about 3 percentage points, increased completed years of schooling by roughly 0.26 years, and lowered the incidence of low-skill occupation employment for rural cohorts by around 5 percentage points, all relative to urban cohorts in the same counties and birth cohorts.

To gauge the economic significance of these effects, it is useful to translate the schooling and occupational changes into a simple earnings metric. Micro estimates for China typically place Mincer returns to schooling on the order of 7–10 percent per additional year of education.[7] Using a conservative 7 percent return, our average 0.26-year narrowing of the rural–urban schooling gap corresponds to roughly a 2 percent reduction in the earnings gap that would arise purely from differences in education. Coupled with a 5 percentage points decline in low-skill occupation employment from a rural baseline well above one-half, this implies a nontrivial reallocation of rural workers away from agriculture, manufacturing, and construction and toward higher-paying, more stable jobs. These back-of-the-envelope calculations are intentionally crude, but they underscore that the effects we estimate are economically sizable, not merely statistically significant.

[7]For example, returns in the 7–10 percent range per additional year of schooling are reported by Li, Luo, and Sicular (2014) using CHNS data and by Heckman and Li (2004) using urban household survey data. The calculation here is purely illustrative and does not rely on any single study.

## 5.4 Geographic exposure: County-level pre-policy capacity of the academic high school

Figure 5 presents descriptive patterns of upper-secondary outcomes by terciles of pre-policy academic high school capacity. Panel A plots academic high school completion rates by birth cohort for counties in the low-, medium-, and high-supply terciles. We use pre-reform academic high school completion as a proxy for academic high school capacity. Panel B plots the corresponding rural–urban gaps in academic high school completion for the same three groups of counties. Two stylized facts emerge.

First, upper-secondary outcomes evolve smoothly across cohorts within each tercile. Both overall completion rates and rural–urban gaps change gradually rather than exhibiting any sharp break around the cohorts exposed to the 1999 expansion of university places. This visual evidence points to slow, locally administered adjustments in academic high school enrollment rather than a discrete, centrally coordinated supply shock at the upper-secondary level tied mechanically to the university expansion.

Second, the evolution of levels and gaps is systematically heterogeneous across terciles, and in opposite directions. On the level side (Panel A), counties with low pre-policy academic high school supply exhibit the largest increases in overall completion (blue line); those with medium supply show moderate increases (green line), and those with high supply display the smallest changes (red line). On the gap side (Panel B), however, the ordering reverses. In the lowest-supply counties, the rural–urban gap barely narrows and even widens slightly. This occurs despite the fact that these counties experience the largest post-1999 expansions in academic high school capacity. Medium-supply counties exhibit some convergence, but the strongest rural catch-up occurs in high-supply counties, where overall completion changes little over time. This pattern suggests that in initially capacity-constrained counties, the additional seats created by the expansion were mainly taken up by urban students, whereas in already well-supplied counties the reform primarily relaxed admission bottlenecks for rural students.

Taken together, these descriptive patterns provide indirect evidence that simple expansion of academic high school capacity per se is not what drives changes in the rural–urban gap in academic high school completion. If capacity growth were the main driver, we would expect the largest rural catch-up precisely in counties where academic high school places expanded the most, yet the strongest convergence is instead observed in already high-supply counties where overall capacity hardly changes. This intuition underpins the more formal heterogeneity analysis that follows, which points to the same conclusion.

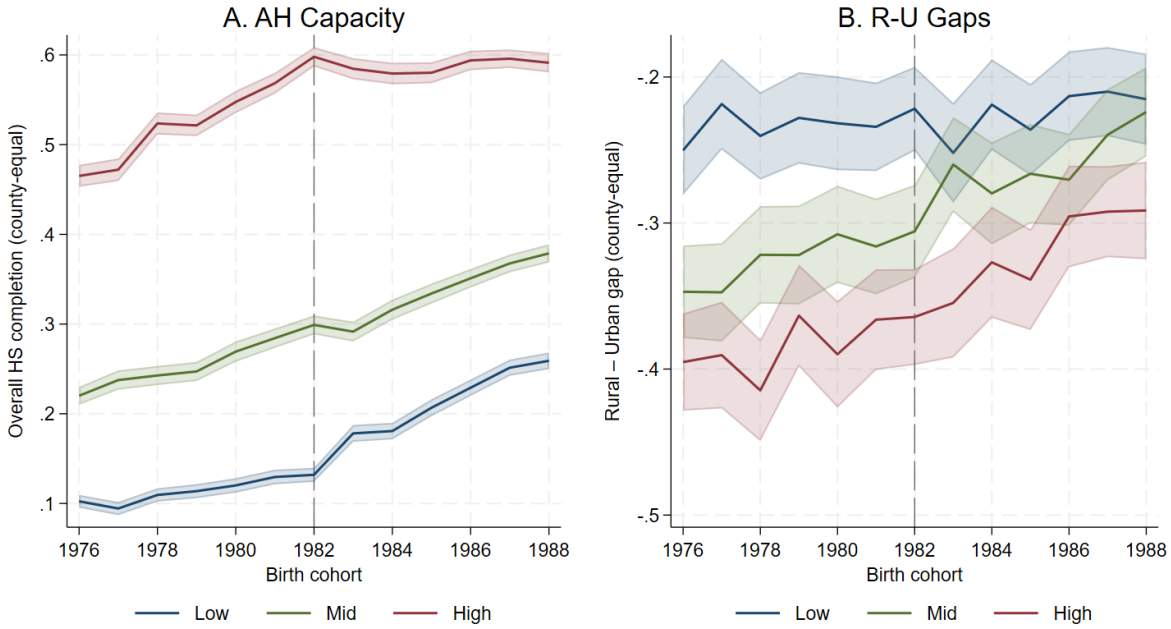


Figure 5: Completion by cohort: levels and rural–urban gaps, by pre-policy capacity
*Notes:* Counties grouped into terciles of pre-policy capacity. Panel A: overall completion (county-equal). Panel B: within-county rural−urban gap (rural minus urban). Shaded bands: 95% CIs across counties.

Table 5 reports estimates of equation (3). In the pooled sample, the coefficient on Rural × Post is about 0.018, implying a post-reform rural–urban difference in academic high school completion of roughly 1.8 percentage points for counties with average pre-policy Capacity. The response varies systematically with pre-policy capacity: the interaction coefficient $\widehat{\delta} = 0.0174$ on Rural × Post × Capacity implies that a one–standard–deviation increase in Capacity raises the post-reform rural gain by about 1.7 percentage points. In our data, this means that the Rural×Post effect is essentially zero in low-capacity counties (one standard deviation below the mean), around 1.8 percentage points at average capacity, and about 3.5 percentage points in high-capacity counties (one standard deviation above the mean). Gender-specific estimates are consistent with this pattern: evaluated at mean Capacity, the rural post-reform gain is about 2.3 percentage points for men and 1.4 percentage points for women, and the slope with respect to Capacity is of similar magnitude across genders (roughly 1.6 and 1.9 percentage points per standard deviation for men and women, respectively).

The cross-panel ordering in Figure 5 and the positive Rural×Post×Capacity estimates in Table 5 are hard to reconcile with a purely seat-driven account of rural convergence. Upper-secondary capacity grows most in counties that start with low academic high school supply and least in counties that are already well supplied, yet the rural–urban gap narrows most in the latter group, where subsequent capacity growth is modest. A

more natural interpretation is a behavioral-response mechanism operating under heterogeneous local constraints: the expansion of university places raises expected returns to the academic track, but rural students can translate these higher returns into academic high school enrollment only where pre-policy upper-secondary supply is relatively slack, whereas in low-supply counties tighter capacity and stronger urban competition limit the scope for rural gains despite larger incremental seat expansions. In this sense, the key predictor of rural convergence is the pre-policy looseness of upper-secondary admissions, not contemporaneous growth in upper-secondary seats.

Table 5: Rural–urban post-reform differences scale with pre-policy capacity

| | **Completed Academic High School** | | |
|---|---|---|---|
| | **Pooled** | **Male** | **Female** |
| Rural × Post | 0.018*** | 0.023*** | 0.014*** |
| | (0.004) | (0.006) | (0.005) |
| Rural × Post × Capacity | 0.017*** | 0.016*** | 0.019*** |
| | (0.004) | (0.005) | (0.005) |
| County and cohort FE | Yes | Yes | Yes |
| County×Rural FE | Yes | Yes | Yes |
| Clustering level | County | County | County |
| Observations | 274,569 | 139,871 | 134,429 |

*Notes*: Post indicates cohorts born 1983 or later. Rural denotes rural registration (as defined in the data section). Capacity is the county-level, standardized pre-policy (cohorts 1976–1982) academic-high-school completion rate and is used to characterize heterogeneity. Thus, the coefficient on Rural × Post × Capacity captures how the post-reform rural–urban difference varies with a one-s.d. higher pre-policy capacity. All specifications include standard demographics and parental education and absorb county FE, cohort FE, and county×registration FE. Robust standard errors are clustered at the county level.

# 6 Robustness checks

In this section, we examine the robustness of the main Rural × Post estimates to a variety of alternative specifications and sample definitions. First, we document that there is no discrete jump in academic high school enrollment around 1999, consistent with our interpretation that the reform primarily operates through changes in the returns to the academic track rather than a contemporaneous expansion of upper-secondary seats. We then show that rural–urban gaps in academic high school completion exhibit flat pre-trends prior to the reform, alleviating concerns about differential convergence that

predates the expansion. Next, we verify that our results are stable when we redefine rural and urban status based on a more restrictive sample of non-migrant individuals and when we drop boundary cohorts that may only be partially exposed to the reform. Finally, we demonstrate that the estimated effects are robust to using an alternative definition of low-skill employment and to clustering standard errors at different levels, indicating that our conclusions are not sensitive to the precise outcome definition or inference procedure.

## 6.1 Academic high school enrollment growth around the 1999 reform

To assess whether our results could be driven by a contemporaneous discrete expansion of academic high school capacity at the upper-secondary level, we use aggregate data from the *China Educational Statistics Yearbook*. The yearbook reports annual academic high school enrollments by school location (urban vs. rural) and total higher-education admissions.[8]

Figure A3 plots annual percentage changes in urban and rural academic high school enrollments and in higher-education admissions for 1996–2003. The growth rate of higher-education admissions jumps to almost 50 percent in 1999 and remains elevated in the early 2000s, reflecting the well-known tertiary expansion. By contrast, the growth rates of both urban and rural academic high school enrollments fluctuate in a much narrower band—around 10–20 percent in most years—with no exceptional surge in 1999 and no systematic divergence between urban and rural areas.

The absence of a sharp, rural-specific acceleration in high school enrollment growth around 1999 suggests that there was no discrete expansion of rural academic high school capacity coinciding with the tertiary reform. This supports our interpretation that the main policy-induced break during our study period occurred at the higher-education margin, and that the rural–urban changes we document are unlikely to be driven by a contemporaneous wave of rural upper-secondary expansion.

## 6.2 Regional cohort-level rural–urban gaps in academic high school completion

As a complementary check, we construct academic high school completion rates at the county level separately for rural and urban residents and then track how the rural–urban difference evolves across cohorts. Let $c$ index counties, $y$ index birth cohorts, and $r \in \{\mathrm{R}, \mathrm{U}\}$ denote rural and urban registration. For each cell $(c, y, r)$, I define the

[8] In the yearbook, "urban" and "rural" are defined by the location of the school rather than students' household registration status, so the series capture the evolution of upper-secondary *supply* in urban and rural areas.

completion rate

$$\mathrm{HS}_{cyr} = \frac{1}{N_{cyr}} \sum_{i \in (c,y,r)} \mathbf{1}\{\text{Completed AH}_i = 1\},$$

where $N_{cyr}$ is the number of individuals in county $c$, cohort $y$, and hukou group $r$. I then form the county–cohort rural–urban gap in academic high school completion as

$$\mathrm{gap}_{cy}^{\mathrm{HS}} = \mathrm{HS}_{cy\mathrm{R}} - \mathrm{HS}_{cy\mathrm{U}},$$

and keep only cells for which both rural and urban completion rates are observed. In the regression analysis, each county–cohort cell is weighted by its total sample size $N_{cy} = N_{cy\mathrm{R}} + N_{cy\mathrm{U}}$, so that gaps estimated from larger underlying samples receive more weight than gaps based on only a few observations. County fixed effects are included to absorb time-invariant differences in rural–urban gaps across counties.

We regress the county–cohort gap $\mathrm{gap}_{cy}^{\mathrm{HS}}$ on a full set of birth-year indicators with the 1982 cohort as the omitted category, controlling for county fixed effects and clustering standard errors at the county level. The constant term can be read as the average rural–urban gap in academic high school completion for the 1982 cohort, while each birth-year coefficient measures how the gap for that cohort differs from the 1982 gap. The estimates imply that, for the 1982 cohort, rural residents complete academic high school roughly 29 percentage points less often than urban residents in the same county. For pre-1982 cohorts, the coefficients are small and imprecise, suggesting no strong systematic pre-trend in the rural–urban gap. From the mid-1980s cohorts onward, the coefficients turn positive and increase in magnitude: the gap is around 2–3 percentage points smaller than in 1982 for the 1984–1986 cohorts, and 5–6 percentage points smaller for the 1987 and 1988 cohorts. Overall, the county–cohort evidence points to a gradual narrowing of the rural–urban gap in academic high school completion, in line with the individual-level results.

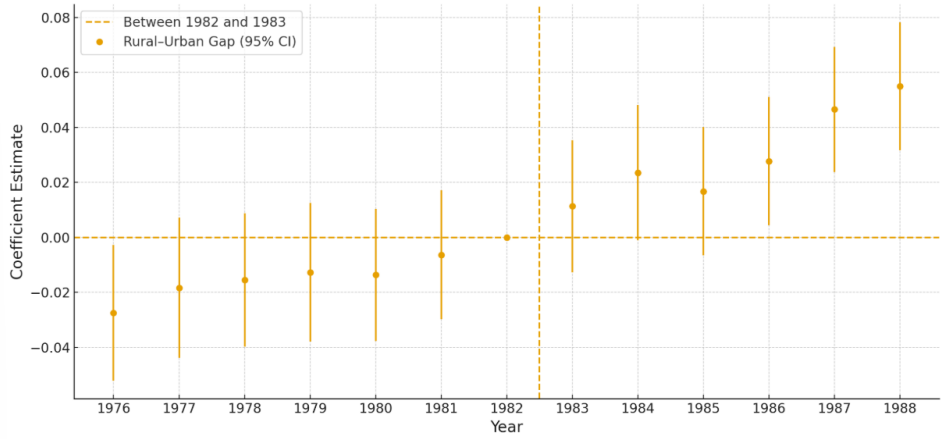


Figure 6: County–cohort rural–urban gap in academic high school completion

Notes: Each point plots the coefficient on a birth-year indicator from a regression of the county–cohort rural–urban gap in academic high school completion on birth-year dummies, with 1982 as the omitted category. The dependent variable is defined as the completion rate among rural residents minus the completion rate among urban residents in the same county and birth cohort. The regression includes county fixed effects and is weighted by the total number of individuals in each county–cohort cell. Vertical bars denote 95 percent confidence intervals. Standard errors are clustered at the county level.

## 6.3 Robustness to Hukou-Based Sample Definition

A key assumption underlying the identification strategy is that individuals' rural hukou status remains stable over the study period and is not systematically affected by the implementation of the higher education expansion policy. Under China's hukou system, conversion from urban to rural hukou is virtually impossible, which eliminates concerns over misclassification due to "hukou downgrading." While it is theoretically possible for individuals to convert from rural to urban hukou, such transitions were difficult to achieve in the prevailing institutional and social environment at the time.

Moreover, rural hukou status has long been associated with structural disadvantages in access to public services, such as education, healthcare, and employment opportunities. From the perspective of individual preferences, rural hukou is not an attractive status. Within this institutional and social context, individuals have little incentive to voluntarily switch from urban to rural hukou, even in response to improvements in compulsory education.

In this study, sample assignment is based on hukou status as recorded in the 2015 survey and includes individuals holding either rural or urban hukou as of that year. If some rural hukou holders benefited from the reform and subsequently converted to urban hukou through higher education attainment, urban employment, or household registration changes, such policy-induced transitions may result in the selective exclusion of individuals who experienced the greatest policy gains. Consequently, our estimates are likely conservative and reflect a lower bound of the true effects of the policy.

While individual hukou conversions are not directly observed in our data, we also re-estimate the baseline on a restricted sample that excludes individuals who have moved away from their place of hukou registration. Specifically, we drop observations whose 2015 place of residence differs from their registered county, thereby removing the subset most likely to have left their registration county in response to improved educational and labor-market opportunities. As shown in Table A6, estimates in this restricted sample remain strong and, if anything, are slightly larger than in the Pooled sample, consistent with the view that baseline effects are conservative when potential rural upgraders are misclassified as urban in 2015.

## 6.4 Robustness: Excluding Urban Districts

A natural concern is that, although urban districts and ordinary counties are both classified as county-level units, their economic structures differ sharply, so they are likely to respond very differently to the reform. In China's administrative hierarchy, prefecture-level cities contain an urban core (the "urban district") as well as surrounding counties and county-level cities; the urban districts are typically the least agriculturally dependent units, with dense service and tertiary sectors and unusually strong access to education. To address this concern, Table A7 reports estimates of the main difference-in-differences specification on a sample that excludes all urban-district counties. The resulting Rural×PostCohort coefficients remain positive, precisely estimated, and economically meaningful: the effect on years of schooling is 0.242 (versus 0.255 in the Pooled sample), the effect on academic high-school completion is 0.027 (versus 0.029), and the effect on low-skill employment is $-0.046$ (versus $-0.051$). In all cases the estimates change little, indicating that the main findings are not an artifact of including urban cores.

The modest attenuation of the coefficients when urban districts are removed is also consistent with the heterogeneity patterns documented in Table 8. There we showed that rural gains from the expansion are smaller in provinces with heavier agricultural dependence and larger in more diversified local economies. Urban-district counties are, by construction, among the least agriculturally dependent units; excluding them mechanically shifts the sample toward counties with higher agricultural dependence, where our heterogeneity analysis already predicts weaker rural gains. The slightly smaller Rural×PostCohort effects in Table A7 therefore reinforce, rather than qualify, the interpretation that the reform delivers the largest benefits to rural students in areas that are less dependent on agriculture and better positioned to translate improved educational opportunities into non-agricultural jobs.

## 6.5 Placebo Test with Pre-Reform Cohorts

As a falsification exercise, we conduct a placebo test using only cohorts born between 1976 and 1982, who were not fully exposed to the 1999 university expansion. These cohorts would have already completed most or all of their high school education by the time the policy was implemented, and should not have altered their education or employment trajectories in response to the reform.

We define a placebo treatment indicator for individuals born in 1981 or 1982, assuming that if our identification strategy is valid, no significant rural treatment effect should be observed in this restricted sample. Table A9 reports the results. The estimated rural-by-post interaction coefficients are small in magnitude and statistically indistinguishable from zero across three outcomes: low-skill employment, years of schooling, and the probability of completing high school.

These findings support the validity of our identification strategy by showing that treatment effects do not appear in cohorts that could not plausibly respond to the policy. Moreover, they help rule out mechanical time trends or cohort composition shifts as potential explanations for our main results.

## 6.6 Excluding Border Cohorts

The baseline specification pools all cohorts born between 1976 and 1988, treating those born in 1983 and later as fully exposed to the 1999 university expansion. A potential concern is that the two earliest treated cohorts (1983–1984) may still be only partially affected by the policy. To evaluate whether our findings hinge on these "border" cohorts, we re–estimate Equation (2) after dropping all individuals born in 1983–1984.

Appendix Table A10 shows that the rural–post interaction remains large and statistically significant. Specifically, excluding the border cohorts yields a 3.8 percentage–point reduction in rural low–skill employment, a 0.34 year increase in completed schooling, and a 5.1 percentage–point rise in the probability of finishing high school. The magnitudes are close to the baseline estimates reported in Table 3, confirming that our core results are not driven by the earliest treated cohorts. Hence, the treatment effect we estimate reflects the broader impact of the expansion rather than idiosyncratic features of the 1983–1984 cohorts.

## 6.7 Alternative Definition of Low-Skill Employment

Our baseline measure of low-skill employment includes agriculture, manufacturing and construction. To verify that the estimated rural effect is not sensitive to this coding choice, we replicate the analysis using a "narrower definition" that counts only agriculture and construction as low-skill jobs. This classification captures the two sectors most commonly

associated with manual, low-education tasks in rural China.

Appendix Table A8 reports the results. The coefficient on the rural–post interaction remains negative and highly significant: a 4.5 percentage-point reduction in the likelihood that rural individuals work in narrowly defined low-skill sectors, nearly identical in magnitude to the baseline estimate. The stability of the effect under this more restrictive outcome definition confirms that our main findings are not driven by the particular way low-skill employment is classified.

## 6.8 Robustness to Alternative Clustering Levels

Inference can be sensitive to how residual correlation is modeled. In our context, shocks may be shared (i) *within county–cohort cells*—students from the same county and birth cohort face common secondary-school environments—and (ii) *across counties within the same prefecture*, where a small share of students moves across county lines. We report two complementary clustering schemes: clustering at the county by cohort cell level and one–way clustering at the city (prefecture) level.

Table A11 shows that the point estimates for Rural × Post are unchanged across schemes, and statistical significance is stable. County×cohort clustering yields slightly tighter standard errors, while city clustering is marginally more conservative; in both cases $t$-statistics remain comfortably above conventional thresholds. Overall, the main conclusions do not depend on the clustering level.

# 7 Confounding Factors

Robustness to concurrent shocks is crucial for a causal reading of our results. We re-estimate the main specification by adding triple-difference terms that interact each confounding shock with the rural indicator and the post-reform cohort indicator. These interactions allow the effects of concurrent events to differ for rural cohorts in the post-reform period. While the point estimates shift in some specifications, the core rural–post effect remains economically meaningful and our qualitative conclusions are unchanged. Overall, the results do not appear to be driven by concurrent shocks.

## 7.1 Shock-Driven Confounders

### 7.1.1 Robustness to Macroeconomic Crisis Exposure

To further assess robustness, we examine whether the estimated policy effects are confounded by concurrent macroeconomic shocks, in particular the 1998 Asian Financial Crisis. The crisis hit coastal and export-oriented provinces harder, potentially depressing household incomes and schooling investments, especially in rural areas. We classify

provinces as "high crisis exposure" if their 1997 export-to-GDP ratio exceeded 20% or their FDI share exceeded 5%. We then estimate a triple-difference specification that interacts rural status, post-reform cohort, and an indicator for high crisis exposure.

Table A12 reports the results. The rural-by-post coefficient remains positive and statistically significant for years of schooling (0.326) and high school completion (0.039), and negative and significant for low-skill employment ($-0.069$). The triple interaction is small and not statistically significant for years of schooling ($-0.092$) and high school completion ($-0.013$), but positive and significant for low-skill employment (0.029). These patterns suggest that provinces more exposed to the crisis experience a slightly weaker decline in low-skill employment, but there is little evidence that crisis exposure systematically moderates the schooling gains associated with the 1999 education expansion.

### 7.1.2 Robustness to Natural Disaster Exposure: The 1998 Yangtze Floods

In addition to macroeconomic shocks, we consider whether concurrent natural disasters may confound our estimated effects. The 1998 Yangtze River floods, among the most severe natural disasters in recent Chinese history, disproportionately affected rural areas in provinces such as Hubei, Hunan, Jiangxi, and Anhui. We construct an indicator for counties officially designated as flood-affected in 1998 and estimate a triple-difference model that interacts rural status, post-reform cohort, and flood exposure.

Table A13 shows that the rural-by-post coefficient remains significantly positive for years of schooling (about 0.3 years) and high school completion (about 2.9 percentage points), and significantly negative for low-skill employment (about 4.6 percentage points). By contrast, the triple interaction with flood exposure is close to zero and statistically insignificant for the two schooling outcomes (around -0.1 years and -1.0 percentage points), but positive and significant for low-skill employment (about 5.0 percentage points). These results suggest that the main gains in rural schooling are not driven by differential exposure to the 1998 floods, although crisis-hit areas may have experienced a modest dampening of the reform's impact on shifting workers out of low-skill jobs.

## 7.2 Policy-Related Confounders

### 7.2.1 Do Nine-Year Campaign Counties Alter the Reform Effect?

The Nine-Year Compulsory Education Campaign (1998–2000) targeted underdeveloped counties to expand basic education. To assess whether this campaign interferes with our estimates of the 1999 higher education expansion, we estimate a triple-difference specification that interacts rural status, post-reform cohort, and a county-level campaign indicator. Table A14 shows in non-campaign counties, the rural-by-post coefficient implies that the expansion raises rural years of schooling by about 0.3 years, increases the

probability of completing academic high school by roughly 2.9 percentage points, and reduces the probability of working in low-skill jobs by about 4.2 percentage points.

The triple interaction with the campaign indicator is small and statistically imprecise for years of schooling (about 0.1 years) and for high school completion (about 2.1 percentage points), but is moderately negative for low-skill employment (about 2.6 percentage points, significant at the 10 percent level). This pattern suggests that participation in the Nine-Year Campaign does not overturn the main rural gains from the 1999 expansion. If anything, campaign counties exhibit a slightly larger reduction in low-skill employment, consistent with the idea that earlier improvements in basic schooling may have helped rural cohorts convert the expanded higher-education opportunities into better occupational outcomes.

### 7.2.2 Assessing Confounding from SOE Restructuring

To examine whether the estimated policy effects are contaminated by concurrent structural changes in the labor market, we focus on the large-scale layoffs from state-owned enterprises during China's restructuring wave and exploit cross-province variation in SOE exposure. Specifically, we interact rural status, post-reform birth cohort, and a standardized measure of the 2000 provincial share of urban employment in SOEs, and estimate a triple-difference specification with county, cohort, and county-by-rural fixed effects.

Table A15 reports the results. The rural-by-post coefficient is positive and highly significant for years of schooling (about 0.3 years) and high school completion (about 3.1 percentage points), and negative and significant for low-skill employment (about 4.5 percentage points). The triple interactions with SOE exposure are negative and statistically significant for years of schooling and high school completion, but the magnitudes are small (on the order of 0.1 years of schooling and about 1 percentage point in completion for a one-standard-deviation increase in SOE share), and close to zero and insignificant for low-skill employment. Overall, we do not see strong evidence that the core rural gains from the 1999 expansion are driven by, or highly sensitive to, provincial variation in SOE restructuring; if anything, exposure to SOE-heavy provinces slightly dampens the schooling response without overturning the main patterns.

### 7.2.3 Controlling for the China Trade Shock

To ensure that the estimated effect of the higher education expansion is not confounded by concurrent macroeconomic trends, we control for provincial exposure to the China trade shock following the country's accession to the World Trade Organization (WTO) in 2001. The WTO entry triggered significant structural changes through increased export competition, industrial reallocation, and labor market adjustments. These transformations could have affected household educational investments, particularly in rural regions.

To capture this, we construct a continuous measure of provincial trade exposure based on pre-accession trade intensity, measured in thousand USD per capita. We then estimate a triple-difference specification that interacts rural status, post-reform cohort, and trade exposure.

Table A16 shows that the reform increased the probability of completing high school by 2.2 percentage points for rural students born after the policy. The triple interaction with WTO exposure is small and statistically insignificant across all three outcome variables, suggesting that trade exposure does not moderate the policy's effects. These results provide additional evidence that the observed policy impact is not driven by concurrent globalization shocks.

# 8 Heterogeneity Analysis

## 8.1 Heterogeneous Effects by Expansion Intensity

Table 6 reports triple-difference estimates by provincial expansion terciles, with low-expansion provinces as the reference group. In low-expansion provinces, the Rural×PostCohort coefficient implies that the reform increases rural years of schooling by about 0.16 years, while the effect on completing academic high school is small and imprecisely estimated (around 1.1 percentage points). The interaction terms show that rural schooling gains are larger in provinces that expanded college places more aggressively. Adding the interaction coefficients, the implied effects on years of schooling are approximately 0.29 years in medium-expansion provinces and 0.35 years in high-expansion provinces. For academic high-school completion, the corresponding implied effects are about 4.2 percentage points in medium-expansion provinces and 3.5 percentage points in high-expansion provinces, with the medium-expansion effect precisely estimated and the high-expansion effect marginally significant.

These patterns suggest that provincial expansion intensity mainly scales the strength of the rural schooling response, rather than changing its sign or qualitative nature. Where college supply increased more, rural cohorts are more likely to remain in school longer and to complete the academic track, consistent with the idea that larger increases in tertiary places generate stronger perceived returns to upper-secondary schooling. At the same time, the differences across terciles are moderate in magnitude, indicating that the 1999 expansion raised rural educational attainment broadly across provinces, with somewhat larger gains in those that expanded more aggressively.

Table 6: Heterogeneous Effects of Higher Education Expansion by Provincial Expansion Intensity

| | (1) Years of Schooling | (2) Completed AH | (3) Low-Skill Job |
|---|---|---|---|
| Rural × PostCohort | | | |
| | 0.160*** | 0.011 | -0.066*** |
| | (0.043) | (0.007) | (0.007) |
| Rural × PostCohort × Medium Expansion | | | |
| | 0.131** | 0.031*** | 0.027*** |
| | (0.054) | (0.009) | (0.009) |
| Rural × PostCohort × High Expansion | | | |
| | 0.188** | 0.024* | 0.030** |
| | (0.078) | (0.013) | (0.012) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 274,576 | 274,576 | 274,576 |
| $R^2$ | 0.361 | 0.251 | 0.142 |

*Notes::* Triple-difference regressions by provincial expansion-intensity terciles (reference group: low expansion); expansion intensity is defined in Appendix Table A4. All specifications include county and cohort fixed effects and county-by-rural fixed effects, as indicated in the table. Controls include gender, minority status, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * p < 0.10, ** p < 0.05, *** p < 0.01.

## 8.2 Heterogeneity by University Access

**Construction of Access** We proxy provincial university access using the *Chinese Higher Education Density Index* (CHEDI), defined as the number of accredited degree-granting higher-education institutions per million residents. Let $p$ index provinces and $t$ years. We compute the 1998 cross-sectional density as

$$\text{CHEDI}_{p,1999} = \frac{\text{HEICount}_{p,1998}}{\text{Population}_{p,1998}} \times 10^6,$$

where $\text{HEICount}_{p,1998}$ counts Ministry-of-Education–registered, degree-granting HEIs located in province $p$ in 1999, and $\text{Population}_{p,1998}$ is the resident population of province $p$

in 1999. Our heterogeneity regressor is the standardized index

$$\text{Access}_p \equiv \frac{\text{CHEDI}_{p,1998} - \overline{\text{CHEDI}_{\cdot,1998}}}{\text{sd}\Big(\text{CHEDI}_{\cdot,1998}\Big)},$$

so that $\beta_{RPA}$ is interpretable as the change in the Rural×Post effect associated with a one–standard-deviation increase in pre-reform university density.

We test whether provincial university access moderates the reform's impact on rural–urban gaps by interacting the DID term with a standardized index of university density from the Chinese Higher Education Density Index (CHEDI). Specifically, we estimate

$$Y_{icy} \;=\; \cdots \;+\; \beta_{RP}\Big(\text{Rural}_i \times \text{Post}_y\Big) \;+\; \beta_{RPA}\Big(\text{Rural}_i \times \text{Post}_y \times \text{Access}_p\Big) \;+\; \varepsilon_{icy}, \quad (3)$$

absorbing county, cohort and county×rural fixed effects and clustering at the county level. Because *Access* varies at the province level and the fixed-effects structure removes location and cohort main effects, the *Access* main effect and its lower-order interactions are absorbed. Hence, $\beta_{RPA}$ measures how the Rural×Post effect changes with a one-standard-deviation increase in provincial university access.

Table 7 presents two measures of access: (i) *Total CHEDI* (all tertiary programs) and (ii) *CHEDI-4yr* (four-year institutions). The average DID remains stable and significant: Rural×Post raises academic high school completion by about 0.03 (3 percentage points), increases years of schooling by around 0.26 years, and reduces low-skill employment by roughly 0.05 (5 percentage points).

Under Total CHEDI (Panel A), the triple interaction is small and statistically insignificant across outcomes: provincial university density, when measured by all tertiary places, does not meaningfully change the Rural×Post effect. By contrast, under CHEDI-4yr (Panel B), provincial access to four-year universities does strengthen the schooling response. A one-standard-deviation higher four-year university access is associated with roughly 0.009 more Rural×Post effect on academic high school completion (about 30% of the baseline 0.030) and 0.074 additional years of schooling (about 28% of the baseline 0.262), both statistically significant. For low-skill employment, the modulation remains close to zero and is statistically insignificant.

Overall, these results indicate that provinces with denser four-year university systems see a stronger rural schooling response to the 1999 expansion, even though the average Rural×Post effect is present throughout the country. This is consistent with a two-stage interpretation of the schooling sequence: the reform shifts incentives at the upper-secondary track choice margin everywhere, and this shift translates into particularly large gains in settings where rural students face better access to four-year universities.

Table 7: Does University Access Modulate the Reform Effect? Rural×Post×Access

| | (1) Years of Schooling | (2) Completed AH | (3) Low-Skill Job |
|---|---|---|---|
| **Panel A: Total CHEDI, standardized** | | | |
| Rural × Post | 0.029*** | 0.260*** | -0.046*** |
| | (0.004) | (0.024) | (0.004) |
| Rural × Post × Access ($z$) | 0.005 | 0.043 | -0.001 |
| | (0.004) | (0.026) | (0.004) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 274,576 | 274,576 | 274,576 |
| **Panel B: CHEDI-4yr, standardized** | | | |
| Rural × Post | 0.030*** | 0.262*** | -0.046*** |
| | (0.004) | (0.024) | (0.004) |
| Rural × Post × Access4 ($z$) | 0.009** | 0.074*** | -0.002 |
| | (0.004) | (0.025) | (0.004) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 274,576 | 274,576 | 274,576 |

*Notes*:: Each column reports estimates from a separate triple-difference regression. The triple-interaction coefficient captures the change in the Rural×Post effect per one-standard-deviation increase in provincial university access. Fixed effects are as indicated in the table. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The appendix reports wild-cluster bootstrap p-values clustered at the province level and tercile-split robustness checks.

## 8.3 Heterogeneity by Regional Agricultural Dependence

Table 8 presents heterogeneous effects of the 1999 college expansion across counties with different levels of agricultural dependence. We estimate Equation (1) separately for counties in the bottom, middle, and top terciles of the agricultural share of GDP. The three outcomes are years of schooling (Panel A), completion of academic high school (Panel B), and low-skill industry employment (Panel C).

The schooling coefficients display a clear gradient. In low-agriculture counties, the re-

form increases years of schooling by 0.3 years and raises the probability of completing academic high school by 4.9 percentage points. In counties with middle agricultural shares, the corresponding effects are 0.3 years and 2.8 percentage points. In high-agriculture counties, the estimated impacts are 0.2 years and 3.0 percentage points.

Labor market outcomes show a similar pattern. The probability of entering low-skill industries falls by 4.4 percentage points in low-agriculture areas, 5.7 percentage points in mid-agriculture areas, and 4.6 percentage points in high-agriculture areas. These results indicate that local economic structure conditions both the schooling response to the expansion and the extent to which additional education translates into shifts out of low-skill employment.

Overall, the evidence suggests that the average effects of the 1999 expansion mask substantial cross-county heterogeneity by agricultural dependence. In the absence of complementary improvements in local schooling systems and labor demand, national higher-education reforms may yield uneven gains across regions.

Table 8: Heterogeneous Impacts of the 1999 Expansion by Agricultural GDP Share

| | **Low ag. share** | **Mid ag. share** | **High ag. share** |
|---|---|---|---|
| *Panel A. Outcome: Years of Schooling* | | | |
| Rural × PostCohort | 0.347*** | 0.282*** | 0.249*** |
| | (0.044) | (0.045) | (0.045) |
| $R^2$ | 0.336 | 0.307 | 0.267 |
| *Panel B. Outcome: Completed Academic High School* | | | |
| Rural × PostCohort | 0.049*** | 0.028*** | 0.030*** |
| | (0.007) | (0.008) | (0.007) |
| $R^2$ | 0.240 | 0.215 | 0.196 |
| *Panel C. Outcome: Low-Skill Industry Employment* | | | |
| Rural × PostCohort | −0.044*** | −0.057*** | −0.046*** |
| | (0.008) | (0.007) | (0.008) |
| $R^2$ | 0.112 | 0.139 | 0.141 |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 77,908 | 85,961 | 110,707 |

*Notes::* Each coefficient reports the estimated effect of the 1999 expansion from a separate difference-in-differences regression estimated within terciles of the agricultural GDP share (low, middle, high). All specifications include county and cohort fixed effects and county-by-rural fixed effects, as indicated in the table. Controls include gender, minority status, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# 9 Conclusion

This study exploits the sharp nationwide surge in university admissions in 1999 as an exogenous increase in tertiary-level capacity and traces its effects back to the secondary-school tier. Using individual records from the 2015 1 % Population Sample Survey of China and a difference-in-differences design with county, cohort and county-by-rural fixed effects, we find that the expansion raised the probability that rural youths completed the academic high school track by roughly 3 percentage points, increased their average years of schooling by about 0.3 years, and lowered their likelihood of working in low-skill occupations by approximately 5 percentage points. Placebo tests, event-study diagnostics, and an extensive battery of robustness checks, together with targeted tests for confound-

ing factors, support a causal interpretation. Heterogeneity analysis further shows that the gains are markedly smaller in counties with high agricultural dependence and are somewhat larger in provinces that expanded university capacity more aggressively. In addition, provinces with denser four-year university systems exhibit stronger rural schooling responses to the reform, consistent with the idea that both larger tertiary expansions and better tertiary access amplify upstream adjustments at the academic high school margin.

These findings underscore two broader lessons. First, supply-side reforms at the tertiary tier can meaningfully reshape incentives earlier in the schooling sequence, altering behavior before students reach the reformed level. Second, the equalizing potential of such reforms is neither automatic nor spatially uniform. Regions with thin non-farm labor markets or weaker upper-secondary infrastructure capture smaller gains, while inter-provincial college mobility relaxes local capacity constraints but makes provincial intensity a noisy proxy for actual access.

Future efforts to narrow rural–urban gaps should pair expansion at the tertiary tier with targeted investments earlier in the schooling sequence. Concretely, policymakers could (i) upgrade rural senior-secondary facilities and offer performance-linked stipends to attract and retain qualified teachers, with a particular focus on boarding options that lower commuting costs for remote students; (ii) expand digital infrastructure and blended-learning platforms so that high-quality instruction reaches counties with scarce subject specialists; (iii) pilot conditional cash transfers or tuition-waiver top-ups that defray out-of-pocket costs for low-income households during the junior-to-senior-secondary transition; and (iv) develop school-to-work programs that link academic high schools with local non-farm employers, increasing the likelihood that additional schooling translates into concrete employment prospects. Embedding rigorous monitoring and randomized rollout of such complementary measures would also clarify which mechanisms deliver the highest returns when tertiary capacity expands.

More generally, the Chinese case illustrates how aligning incentives across contiguous stages of the education system is crucial for translating enrollment growth into broad-based human-capital accumulation.

# Data Availability

The individual-level data used in this study come from the *2015 1% Population Sample Survey of China*. Because of confidentiality agreements with the National Bureau of Statistics (NBS), the authors are not permitted to release the raw micro-data. Researchers may apply for access through the NBS data application system (`https://data.stats.gov.cn`). Aggregated data tables and variable definition files generated during the study are available from the corresponding author upon reasonable request.

## Code Availability

All Stata do-files and LaTeX scripts used for data cleaning, analysis, and replication are currently being curated and will be deposited in a public repository once the article is formally published. Researchers who wish to inspect the code during the review process may obtain it from the corresponding author upon reasonable request.

## Declaration of Interests

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, we used OpenAI's ChatGPT in order to improve language clarity and readability. After using this tool, we reviewed and edited the content as needed and take full responsibility for the content of the publication.

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# Online Appendix: Additional Figures and Tables

# 1 Additional Descriptive Statistics

## 1.1 China's Administrative Geography

China's administrative geography is hierarchical. For the purposes of this paper, the relevant structure can be summarized as *province* → *prefecture-level city* → *county-level units*.

At the top level, provinces are the primary subnational jurisdictions. Below provinces, most areas are organized into *prefecture-level cities*. Despite the label "city," a prefecture-level city typically denotes an administrative region that includes both a central urban core and surrounding areas.

Within a prefecture-level city, the next tier consists of county-level units. Two types are especially common. First are *urban districts*, which cover the built-up urban area, adjacent neighborhoods of the city proper and small part of rural area. Second are *counties* and county-level cities, which are generally located outside the central districts and often encompass peri-urban towns and rural areas. As a result, a single prefecture-level "city" in administrative data often contains a mixture of urban districts and surrounding counties.

This distinction is important for interpretation because many public services, including basic and upper-secondary education, are commonly administered and financed at the county level (counties and districts). Accordingly, throughout the paper we treat county-level units as the relevant local administrative unit and cluster statistical inference at that level.

## 1.2 Parental Education Trends

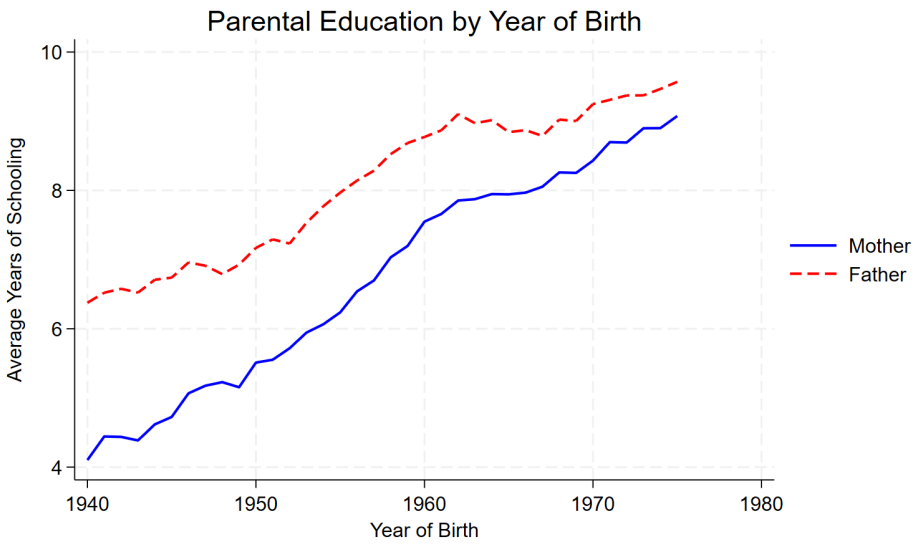



Figure A1: Parental Education by Year of Birth

*Notes*: This figure plots average parental education by child birth year using the 2005 1% Census. Parental education is approximated by county-level mean education of parental cohorts inferred from average age at childbirth.

## 1.3 Schooling stage at the 1999 expansion

Table A1: Schooling stage by birth cohort at the time of the 1999 expansion

| Cohort | Age in 1999 | 1998–1999 school yr. | Track-choice timing | Reform exposure |
|---|---|---|---|---|
| 1981 | 18 | Senior(High) 3/2 | 1997 | No |
| 1982 | 17 | Senior 2/1 | 1998 | No |
| 1983 | 16 | Sen 1/Jun 3 | 1999 | First exposed |
| 1984 | 15 | Junior(Middle) 3/2 | 2000 | Exposed |

*Notes:* Cohort age, schooling stage in the 1998–1999 school year, and reform exposure are determined mechanically from year of birth and the timing of the 1999 expansion; "track choice" refers to the upper-secondary transition.

## 1.4 Trends in Rural and County-Seat High Schools (1985–2005)

Table A2: Evolution of Rural and County High School Counts in China

| Year | Rural HS | County HS |
|---|---|---|
| 1985 | 5,943 | 11,869 |
| 1990 | 4,822 | 10,650 |
| 1995 | 3,112 | 9,000 |
| 2000 | 2,629 | 8,804 |
| 2005 | 2,180 | 9,841 |
| Change 1985–2005 | −63% | −17% |

*Notes:* National totals. "Rural HS" and "County HS" refer to senior high schools located in rural areas and county seats (including both academic and vocational high schools), respectively. The last row reports the percent change between 1985 and 2005. Source: *Educational Statistics Yearbook of China* (various years). This table captures the number of institutions rather than enrollment capacity or seats. While consolidation likely increased average school capacity, it may have reduced geographic accessibility for rural students, potentially raising the effective cost of attending senior high school.

## 1.5 Industry Code Aggregation

Table A3: Concordance Between Three-Digit Industry Codes and Broad Sectors

| Code | Sector |
|---|---|
| 1 | Agriculture, Forestry, Animal Husbandry & Fishery |
| 2 | Mining |
| 3 | Manufacturing |
| 4 | Utilities |
| 5 | Construction |
| 6 | Wholesale/Retail; Transportation/Postal |
| 7 | IT; Finance; Real Estate |
| 8 | Business Services; Research; Education; Health |
| 9 | Culture; Sports; Public Administration |
| 10 | Other Social Organizations |

*Notes*: Aggregation of national three-digit industry codes used to construct the industry categories in the paper.

## 1.6 Provincial-Level Expansion in Higher Education

Table A4: Provincial-Level Changes in Higher Education Supply, 1998–2006

| **Province/Region** | **Proportional Change** |
|---|---|
| Beijing | 0.29 |
| Tianjin | 0.39 |
| Hebei | 0.21 |
| Shanxi | 0.22 |
| Inner Mongolia | 0.14 |
| Liaoning | 0.18 |
| Jilin | 0.17 |
| Heilongjiang | 0.20 |
| Shanghai | 0.24 |
| Jiangsu | 0.23 |
| Zhejiang | 0.23 |
| Anhui | 0.21 |
| Fujian | 0.27 |
| Jiangxi | 0.29 |
| Shandong | 0.20 |
| Henan | 0.22 |
| Hubei | 0.24 |
| Hunan | 0.22 |
| Guangdong | 0.24 |
| Guangxi | 0.19 |
| Hainan | 0.28 |
| Chongqing | 0.34 |
| Sichuan | 0.24 |
| Shaanxi | 0.23 |
| Gansu | 0.16 |
| **National average** | **0.23** |

*Notes::* This table reports, at the provincial level, the average annual increase in the number of university seats between 1998 and 2006, divided by the total number of high school graduates in 1998.

## 1.7 Higher-Education Completion Gap

Table A5: Effect of the 1999 Expansion on Higher-Education Completion

| | **Completed Higher Education** | | |
|---|---|---|---|
| | **Pooled** | **Male** | **Female** |
| Rural × Post | $-0.012^{***}$ | 0.002 | $-0.025^{***}$ |
| | (0.004) | (0.005) | (0.004) |
| Controls | Yes | Yes | Yes |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| $R^2$ | 0.291 | 0.306 | 0.309 |
| Clustering Level | County | County | County |
| Observations | 274,576 | 139,873 | 134,431 |

*Notes*: Each column reports estimates from a separate high-dimensional fixed-effects regression. The dependent variable is an indicator for completing higher education. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include ethnicity and parental education (and sex in the pooled sample, as indicated). Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# 2 Raw Trends and Pre-Treatment Patterns

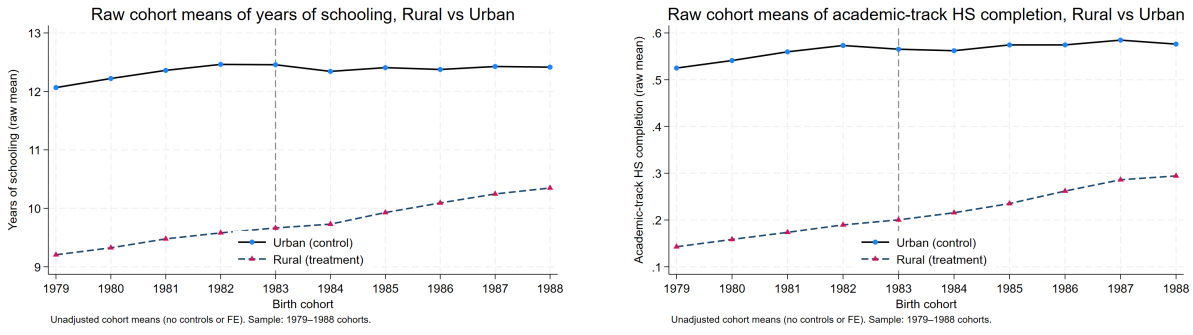
## 2.1 Construction of County-Level Raw Trends

To construct the raw cohort trends, we compute county-by-cohort means separately for rural and urban residents and then take sample-size–weighted averages across counties within each cohort.

The resulting cohort patterns indicate that urban academic high school completion changes relatively little across the post-policy cohorts, while rural completion broadly continues its pre-reform upward trend. These patterns are consistent with the allocation mechanism we emphasize. Following the 1999 college expansion, expected returns to the academic track likely increased for both rural and urban students. However, upper-secondary places remained tightly rationed, and counties were instructed to maintain a stable balance between academic and vocational enrollment. Under these conditions, the reform plausibly affected how a largely fixed stock of academic high school places was allocated across groups, rather than expanding total capacity. Because rural students

started from substantially lower college-going probabilities, the increase in expected returns may have been larger for them, leading to greater competitive pressure from rural students for academic high school seats. As a result, rural completion could rise while the urban series exhibits only modest additional changes.

By contrast, in a counterfactual environment where academic high school capacity were not binding, one might expect both rural and urban completion to increase in tandem over cohorts, with the rural–urban gap remaining relatively stable rather than narrowing, as observed in the data.




(a) Years of schooling (raw)

(b) Academic HS completion (raw)

Figure A2: Raw Cohort Trends: Rural vs. Urban

# 3 Robustness Checks

## 3.1 Robustness: Academic high school enrollment growth around the 1999 reform

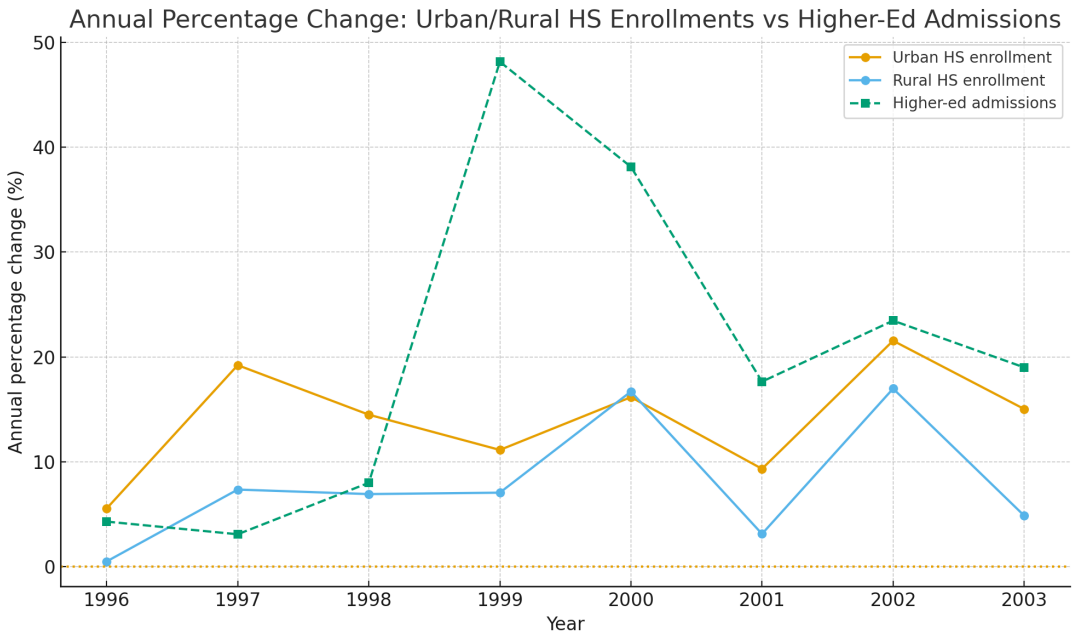



Figure A3: Annual percentage change in urban/rural high school enrollments and higher-education admissions, 1996–2003

*Notes:* The figure plots annual percentage changes in academic high school enrollments for urban and rural schools, together with annual percentage changes in total higher-education admissions. Urban and rural academic high school enrollments are from the *China Educational Statistics Yearbook* and are classified by school location rather than students' household registration status. Higher-education admissions are the number of students admitted through the national college entrance examination. Growth rates are calculated relative to the previous year; 1995 is omitted because no prior year is available.

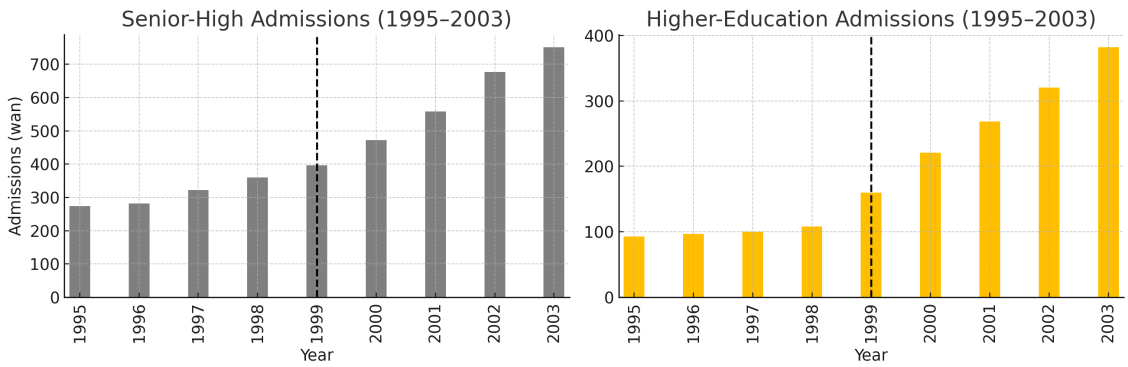



Figure A4: Admissions, 1995–2003: Academic High vs. University

*Notes*: Annual admissions (ten-thousand students). Dashed line marks the policy time.

## 3.2 Robustness: Hukou Transfer

Table A6: Effect of the 1999 Expansion on Education and Occupation (Non-migrant Sample)

| | (1) Years of Schooling | (2) Completed AH | (3) Low-Skill Job |
|---|---|---|---|
| Rural × PostCohort | 0.261*** | 0.030*** | −0.045*** |
| | (0.024) | (0.004) | (0.004) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 273,335 | 273,335 | 273,335 |
| $R^2$ | 0.369 | 0.256 | 0.149 |

*Notes*: This table replicates the baseline specification on a non-migrant sample, excluding individuals whose place of residence differs from their registered hukou county to mitigate bias from policy-induced migration and hukou changes. Fixed effects and controls follow the baseline specification. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## 3.3 Robustness: Excluding Urban Districts

Table A7: Robustness: Excluding Urban Districts

| | (1) Years of Schooling | (2) Completed AH | (3) Low-Skill Job |
|---|---|---|---|
| Rural × PostCohort | 0.242*** | 0.027*** | -0.046*** |
| | (0.025) | (0.004) | (0.004) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 254,902 | 254,902 | 254,902 |

*Notes*: Sample excludes counties classified as urban districts at the administrative-district level (see text for details). The post-expansion period is defined as cohorts reaching upper-secondary age after 1982; the rural group is defined by rural household registration. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, minority status, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## 3.4 Robustness: Alternative Outcome Definitions

Table A8: Robustness: Narrow Definition of Low-Skill Employment

| | (1)<br>Narrow Low-Skill Employment |
|---|---|
| Rural × Post | −0.0447*** |
| | (0.0041) |
| County and Cohort FE | Yes |
| County × Rural FE | Yes |
| Clustering Level | County |
| Observations | 254,597 |

*Notes*: The dependent variable is an indicator equal to one if an individual works in agriculture or construction. The treatment indicator equals one for cohorts born in 1983–1988 and zero for cohorts born in 1976–1982. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## 3.5 Robustness: Placebo Cohorts (1976–82)

Table A9: Placebo Test Using Pre-Reform Cohorts Only (1976–1982)

| | (1)<br>Years of Schooling | (2)<br>Completed Academic HS | (3)<br>Low-Skill Job |
|---|---|---|---|
| Rural × Post | 0.0260 | 0.0070 | −0.0041 |
| | (0.0475) | (0.0087) | (0.0076) |
| County and Cohort FE | Yes | Yes | Yes |
| county × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 138,156 | 138,156 | 138,156 |

*Notes*: Placebo estimates using cohorts born 1976–1982, none of whom were fully exposed to the 1999 expansion. The placebo Post indicator equals one for cohorts born in 1981–1982 and is interacted with the rural indicator. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places.

## 3.6 Robustness: Cohort Window Sensitivity

Table A10: Main DID Estimates (Excluding Border Cohorts)

| | (1) Years of Schooling | (2) Completed Academic HS | (3) Low-Skill Job |
|---|---|---|---|
| Rural × Post | 0.338*** | 0.038*** | −0.051*** |
| | (0.026) | (0.004) | (0.004) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 235,065 | 235,065 | 235,065 |

*Notes*: Estimates from HDFE OLS with county, birth-cohort, and county-by-rural fixed effects. Controls include gender, minority status, father's education, and mother's education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## 3.7 Robustness: Alternative Clustering Levels

Table A11: Robustness to Alternative Clustering Levels

| | (1) Yrs of Schooling | (2) Completed Academic HS | (3) Low–Skill Job |
|---|---|---|---|
| **Panel A: Cluster at County × Cohort** | | | |
| Rural × Post | 0.255*** | 0.029*** | −0.049*** |
| | (0.024) | (0.004) | (0.004) |
| Observations | 274,576 | 274,576 | 274,576 |
| County × Cohort Clusters | 25,947 | 25,947 | 25,947 |
| **Panel B: One–Way Cluster at City Level** | | | |
| Rural × Post | 0.255*** | 0.029*** | −0.049*** |
| | (0.024) | (0.004) | (0.004) |
| Observations | 274,576 | 274,576 | 274,576 |
| City Clusters | 343 | 343 | 343 |

*Notes*: Each column replicates the baseline regression for cohorts born 1976–1988. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, ethnicity, and parental education. Panel A clusters standard errors at the county-by-cohort cell level; Panel B clusters at the city level. Robust standard errors are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# 4 Confounding Factors

## 4.1 1998 Asian Financial Crisis Exposure

Table A12: Heterogeneous Effects of the Education Policy by Crisis Exposure

| | (1) Years of Schooling | (2) Completed Academic HS | (3) Low-Skill Job |
|---|---|---|---|
| *Rural* × *PostCohort* | | | |
| | 0.326*** | 0.039*** | −0.069*** |
| | (0.062) | (0.010) | (0.010) |
| *Rural* × *PostCohort* × *High Crisis Region* | | | |
| | −0.092 | −0.013 | 0.029*** |
| | (0.067) | (0.011) | (0.011) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 274,576 | 274,576 | 274,576 |
| $R^2$ | 0.371 | 0.257 | 0.149 |

*Notes*: Each column reports estimates from a separate triple-difference regression. "High Crisis Region" indicates provinces highly exposed to the 1998 Asian Financial Crisis based on pre-crisis export dependence or FDI share. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## 4.2 1998 Flood Exposure

Table A13: Heterogeneous Effects of the Education Policy by 1998 Flood Exposure

| | (1) Years of Schooling | (2) Completed Academic HS | (3) Low-Skill Job |
|---|---|---|---|
| *Rural* × *PostCohort* | | | |
| | 0.256*** | 0.029*** | −0.046*** |
| | (0.024) | (0.004) | (0.004) |
| *Rural* × *PostCohort* × *Flood98* | | | |
| | −0.125 | −0.010 | 0.050** |
| | (0.154) | (0.026) | (0.025) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 274,576 | 274,576 | 274,576 |
| $R^2$ | 0.437 | 0.339 | 0.243 |

*Notes*: Triple-difference regressions using county-level 1998 flood exposure. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## 4.3 Compulsory Education Campaign

Table A14: Heterogeneous Effects of Education Expansion

| | (1) Years of Schooling | (2) Completed Academic HS | (3) Low-Skill Job |
|---|---|---|---|
| Rural × PostCohort | | | |
| | 0.256*** | 0.029*** | -0.042*** |
| | (0.025) | (0.004) | (0.004) |
| Rural × PostCohort × Campaign | | | |
| | 0.120 | 0.021 | -0.026* |
| | (0.084) | (0.015) | (0.014) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 274,576 | 274,576 | 274,576 |
| $R^2$ | 0.371 | 0.257 | 0.149 |

*Notes*: Each column reports estimates from a separate triple-difference regression. Campaign indicates counties prioritized in the national Nine-Year Compulsory Education Campaign. All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## 4.4 Assessing Confounding from SOE Restructuring

Table A15: Heterogeneous Effects of Education Policy by SOE Exposure

| | (1) Years of Schooling | (2) Completed Academic HS | (3) Low-Skill Job |
|---|---|---|---|
| Rural × PostCohort | | | |
| | 0.263*** | 0.031*** | -0.045*** |
| | (0.024) | (0.004) | (0.004) |
| Rural × PostCohort × SOE Share (2000, standardized) | | | |
| | -0.049** | -0.013*** | -0.004 |
| | (0.024) | (0.004) | (0.004) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 274,576 | 274,576 | 274,576 |
| $R^2$ | 0.371 | 0.258 | 0.149 |

*Notes*: Each column reports estimates from a separate triple-difference regression. SOE exposure is the provincial share of urban employment in SOEs in 2000 (standardized to mean zero and unit standard deviation). All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## 4.5 Controlling for the China Trade Shock

Table A16: Heterogeneous Effects of the Education Policy by WTO Exposure

| | (1) Years of Schooling | (2) Completed Academic HS | (3) Low-Skill Job |
|---|---|---|---|
| *Rural × PostCohort* | | | |
| | 0.248*** | 0.022*** | −0.060*** |
| | (0.045) | (0.008) | (0.007) |
| *Rural × PostCohort × WTO Exposure* | | | |
| | −0.007 | 0.004 | −0.003 |
| | (0.017) | (0.003) | (0.002) |
| County and Cohort FE | Yes | Yes | Yes |
| County × Rural FE | Yes | Yes | Yes |
| Clustering Level | County | County | County |
| Observations | 415,994 | 415,994 | 415,994 |
| $R^2$ | 0.382 | 0.266 | 0.154 |

*Notes*: Triple-difference specification. WTO exposure is measured as provincial per capita exports to WTO members (thousand USD). All specifications include county, birth-cohort, and county-by-rural fixed effects. Controls include gender, ethnicity, and parental education. Robust standard errors clustered at the county level are reported in parentheses. All coefficients and standard errors are reported to three decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# 5 Mechanism: Extended Decomposition

**Mechanism decomposition.** To quantify which schooling margins account for the rural–urban effect on years of schooling, we implement a decomposition that respects the institutional sequencing whereby students must first enter the academic track in upper secondary before they can proceed to tertiary education. We partition individuals into three mutually exclusive tiers: (i) a non-academic tier (at most junior secondary or secondary vocational), (ii) academic high school only, and (iii) tertiary (associate, bachelor, or graduate). Using pre-policy cohorts (1976–1982), I fix tier-specific mean years of schooling and then use four-cell (rural/urban × pre/post) changes in tier shares to decompose the rural–urban difference-in-differences in mean schooling across cohorts 1976–1988.

The six-category version of this decomposition implies a total rural–urban DID in years of schooling of 0.406 years. Of this, Complete academic high school contributes 0.207 years, which corresponds to 50.94% of the total. Progressing to tertiary education

contributes a further 0.054 years, or 13.18% of the total. Together, these two extensive-margin transitions across broad tiers account for roughly two-thirds of the rural gain in schooling.

The remaining 0.146 years (about 35.89% of the total) reflect within-tier re-sorting rather than additional transitions across tiers. The decomposition shows that just over half of the rural gain in years of schooling is driven by new entry into the academic high school program, with a more modest but non-negligible contribution from progression to tertiary education, and the remainder arising from granular reallocation of students within broad schooling tiers.

Table A17: Decomposition of the Rural–Urban DID in Years of Schooling

| Component | Years | Share of six-category DID (%) |
|---|---|---|
| Complete academic HS | 0.207 | 50.94 |
| Progressing to tertiary | 0.054 | 13.18 |
| Within non-academic schooling | 0.160 | 39.49 |
| Within tertiary schooling | -0.015 | -3.61 |
| **Total six-category DID in years of schooling** | **0.406** | **100.000** |

*Notes:* This table decomposes the rural–urban difference-in-differences in mean years of schooling between pre-expansion (1976–1982) and post-expansion (1983–1988) cohorts into contributions from changes in the shares of six mutually exclusive schooling categories ($\leq$ junior secondary, academic high school, vocational high school, associate degree, bachelor degree, graduate degree). Category-specific mean years of schooling are fixed at their pre-expansion cohort averages, and the rural–urban DID is expressed as a weighted sum of changes in category shares.